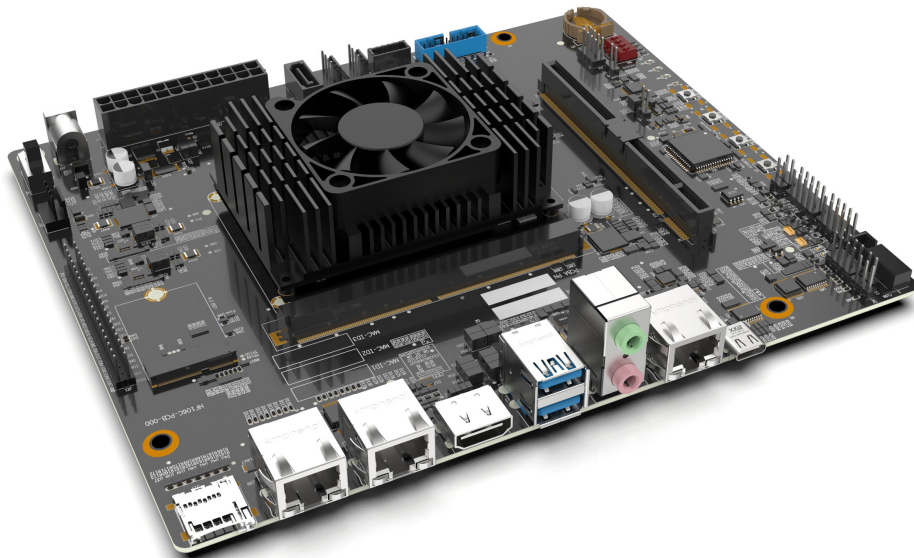


ESWIN

HiFive Premier P550 User Manual

Rev 3.0

2026-3-17



Notice

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Change History

Version	Date	Description
Rev 3.0	2026/03/17	Restructured document structure
Rev 2.3	2025/02/13	Modified product model number
Rev 2.2	2024/09/12	Added SOM image flashing description
Rev 2.1	2024/08/11	Updated user instructions incorporating software user documentation and the P550 customized distribution
Rev 2.0	2024/06/21	Updated User Guide for the new board
Rev 1.0	2024/05/14	Initial version

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1. Safety Notice

1.1. Copyright and Usage Statement

The HiFive_Premier_P550 development board (hereinafter referred to as “P550” , “DVB” , or “Product”) provided by ESWIN is intended for end-user use only, and solely for the purpose of end-user testing and evaluation of ESWIN’ s semiconductor components and software applications (the “Intended Use”). Resale, redistribution, transfer, or any other commercial exploitation of the product is strictly prohibited.

Please note that this document is subject to updates from time to time. You may contact ESWIN sales and after-sales support to obtain the latest version.

1.2. Important Safety Information

1.2.1. Warnings

- Any power supply used with the P550 must comply with the laws, regulations and standards of the country/region where it is used.
- The P550 must be operated in a well-ventilated environment.
- This product must only be used on a flat, non-conductive surface and must never come into contact with conductors.
- Connecting incompatible devices to the P550 may damage the development board.
- All accessories and peripheral devices used with the P550 must comply with the standards of the country/region of use and bear the appropriate certification labels to ensure safety requirements are met.
- All cables and connection equipment used with the P550 must be properly insulated in accordance with safety standards and requirements.

1.2.2. Safety Precautions

The requirements described below must be observed to prevent malfunction and damage to the P550.

- Do not allow the P550 to come into contact with water or expose it to moisture.
- During operation, do not place the P550 on or near conductive surfaces.
- Do not expose the P550 to high temperatures; it may only be used at room temperature and in a ventilated environment as specified in this document.
- Avoid causing any mechanical or electrical damage to the printed circuit board, interfaces, and integrated circuits on the P550.
- Avoid touching, picking up, or moving the P550 while it is powered on.

2. Introduction

The HiFive Premier P550 development board (hereinafter referred to as P550 or DVB) is a high-performance development and evaluation platform designed for next-generation high-performance computing and AIoT applications. It is built upon the EIC7700 AI SoC, the latest offering from ESWIN, which integrates high-performance SiFive RISC-V CPU cores with a powerful AI acceleration unit.

The P550 features an innovative modular design, consisting of two parts: the SOM core board (P550S) and the Carrier board (P550C):

- **P550S (Core Board)**: Integrates a complete minimum system including the EIC7700 SoC, high-speed LPDDR5 memory, eMMC storage, Ethernet PHY, and power management system. It serves as the “brain” of the platform.
- **P550C (Carrier Board)**: Acting as the “skeleton” of the system, it provides stable power delivery, comprehensive external interface expansion, and extensive debugging support.

This design not only ensures the performance and reliability of the core computing unit but also offers extreme interface flexibility and scalability through the carrier board. The platform strictly adheres to Mini-DTX industrial specifications (203mm x 170mm), allowing it to serve as a desktop development powerhouse or be easily integrated into standard 1U server chassis, seamlessly adapting to full-scenario needs from laboratory R&D to edge deployment.

2.1. Hardware Features

The P550 integrates flagship-level hardware configurations within a compact form factor, designed to provide robust support for high-performance computing, AI inference, multimedia processing, and complex networking applications.

Table 1: P550 Key Hardware Specifications

Parameter	Description
Dimensions	203mm x 170mm
CPU	64-bit OoO RISC-V Quad-core Processor
AI Processor	Up to 13.3 TOPS, supporting INT8/INT16/FP16
Memory	On-board 16GB/32GB LPDDR5 @ 6400Mbps
Storage	1x On-board 128GB eMMC
	1x 16MB SPI Flash
	1x SATA 3.0 interface for SSD
	1x Micro SD card slot
	1x On-board 2Kbit EEPROM
Network	2x Gigabit Ethernet RJ45 interfaces
	1x 10/100 Ethernet RJ45 interface
	1x M.2 Key E interface for Wi-Fi module
Display	1x HDMI 2.0 Type A interface supporting 4k@60
	1x MIPI DSI 4-lane FPC interface
Camera	4x MIPI CSI 2-lane or 2x MIPI CSI 2+2-lane
Audio	1x Analog audio interface
	1x F_audio pin header array
PCIe	1x PCI Express Gen3 4-lane via PCIe X16 slot

Table 1: P550 Key Hardware Specifications

Parameter	Description
USB	2x Standard USB 3.2 Gen1 Type-A Interfaces
	1x USB 3.0 19PIN Male Header
	1x Type-E Connector
RTC	1x CR1220 RTC Battery Holder (Battery not included)
Fans	2x 2.54mm 4PIN 12V Fan Interfaces
	1x 2.54mm 3PIN 5V Fan Interface
Buttons	1x SoC Reset Button
	1x Power Button
	1x MCU Reset Button
	1x Recovery Button
Power Supply	1x 24PIN ATX Power Interface or 1x DC 2.5mm Interface (12V, 5A)
Debugging	1x USB Type-C Interface for Debug UART and JTAG
Other I/O	I ² C, I ² S, SPI, UART, General Purpose I/O Ports (mapped to 40-pin header)
Operating Temperature	0~45 °C

2.2. Product Components

The hardware of the P550 development board mainly consists of the SOM core board (P550S) and the Carrier board (P550C). To help you quickly identify key components on the board for subsequent hardware operations and debugging, the figure below provides a numbered location map of the main components, and the table below offers detailed descriptions corresponding to each number.

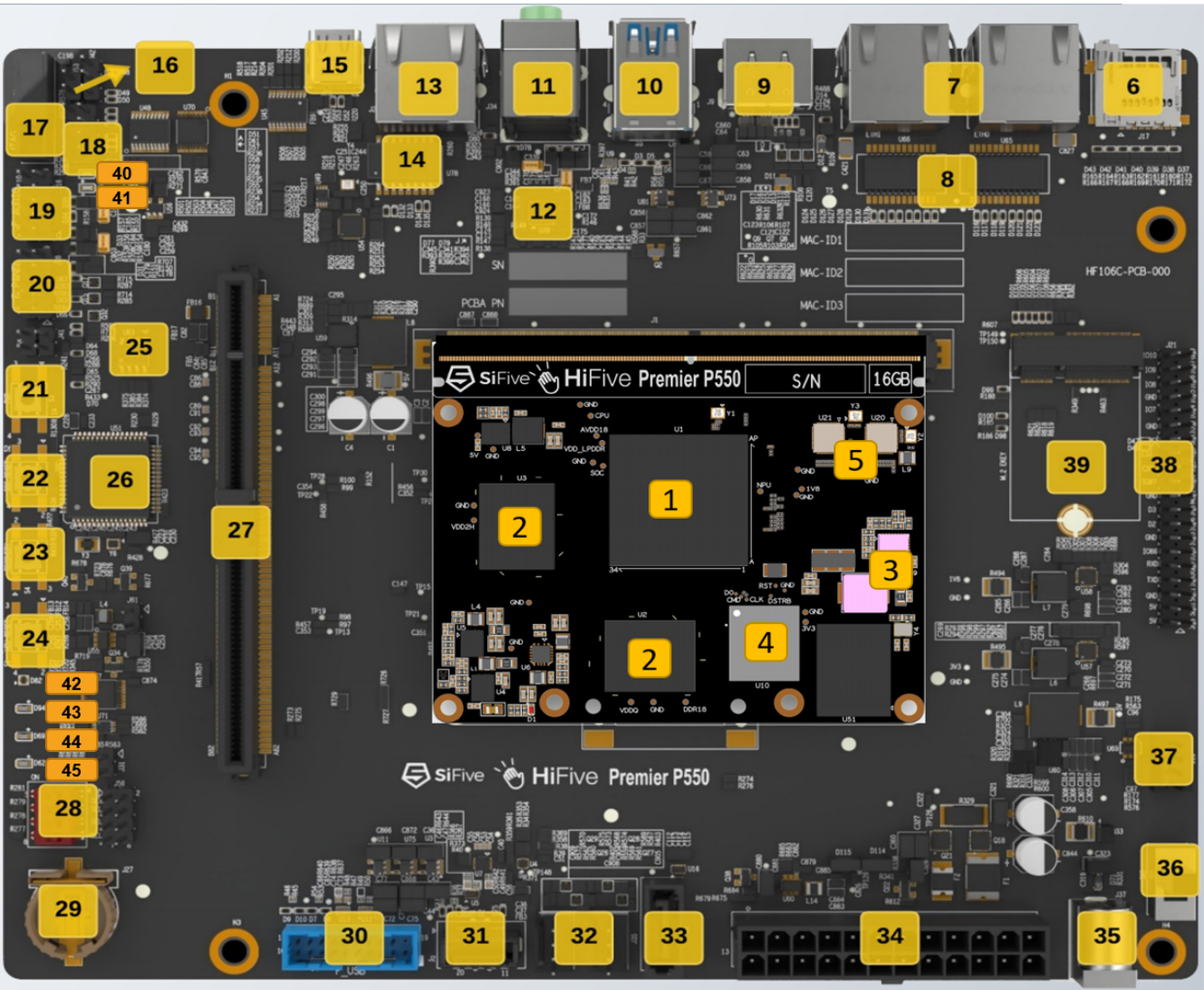


Figure 1: P550 Component Diagram

Table 2: P550 Components

No.	Component Name	No.	Component Name
1	EIC7700 SoC	2	LPDDR5 SDRAM
3	Voltage Regulator / Power Management IC (PMIC)	4	128GB eMMC Flash
5	Ethernet PHY	6	Micro-SD Card Slot
7	SoC Ethernet RJ45 Interface (SoC)	8	Ethernet Transformer
9	HDMI Interface	10	Dual USB Type-A Interfaces
11	Analog Audio Interface	12	Audio Codec
13	MCU Ethernet RJ45 Interface	14	MCU Ethernet Transceiver
15	USB Type-C Interface (Terminal & Debug)	16	SoC JTAG Selector
17	SoC JTAG Interface	18	MCU JTAG Interface

Table 2: P550 Components

No.	Component Name	No.	Component Name
19	Front Panel Audio Interface	20	Front Panel Interface
21	Power Button	22	SoC Reset Button
23	Recovery Button	24	MCU Reset Button
25	Carrier Board EEPROM	26	MCU (Remote Power Control)
27	4-Lane PCIe Connector (x16 Mechanical)	28	Jumper DIP Switch
29	Button Battery Holder	30	USB Type-A Front Panel Interface
31	Type-E Connector for Front Panel Type-C	32	Chassis Fan Interface
33	SATA Interface	34	ATX Power Interface
35	DC 12V Power Adapter Interface (2.5mm)	36	Main Power Switch
37	SoC Fan Interface	38	GPIO Interface
39	M.2 E-Key (SDIO Wi-Fi Module)	40	MCU System Power LED
41	SoC System Power LED	42	RGB LED
43	User-defined LED	44	eMMC LED
45	SATA LED		

2.3. Accessories List

To ensure you can successfully unbox and use the P550 development board, the standard package includes the necessary accessories listed below. Please verify the contents against this list upon receipt.

Item	Accessory Name	Quantity	Remarks
1	P550 Development Board Kit	1	Pre-installed with heatsink and fan
2	USB Type-C Data Cable	1	For connecting the debug interface
3	USB Type-A Data Cable	1	For Fastboot flashing or bootloader recovery
4	HDMI Data Cable	1	For video output
5	Ethernet Cable	1	For network connection
6	Screws and Standoffs	4 each	Screws: M3x5mm; Standoffs: M3x15mm (for fixing)
7	I/O Shield	1	For adapting to standard chassis

2.4. Supporting Software and Documentation

The P550 offers a complete hardware and software support ecosystem. In addition to this “User Manual” which guides hardware usage and basic flashing, please refer to the following resources for in-depth system customization, driver development, and AI application deployment:

- **BSP Quick Start Guide:** Guides you on setting up a cross-compilation environment, customizing the Linux kernel, building root file systems, and compiling the complete SDK.
- **ESSDK Quick Start Guide:** Details the structure and usage of the ESWIN SDK, as well as how to perform AI model quantization, compilation, and on-board inference deployment.

You can obtain the latest documentation, tools, and images through ESWIN' s official technical support channels.

3. Design Diagram

3.1. EIC7700 Chip Block Diagram

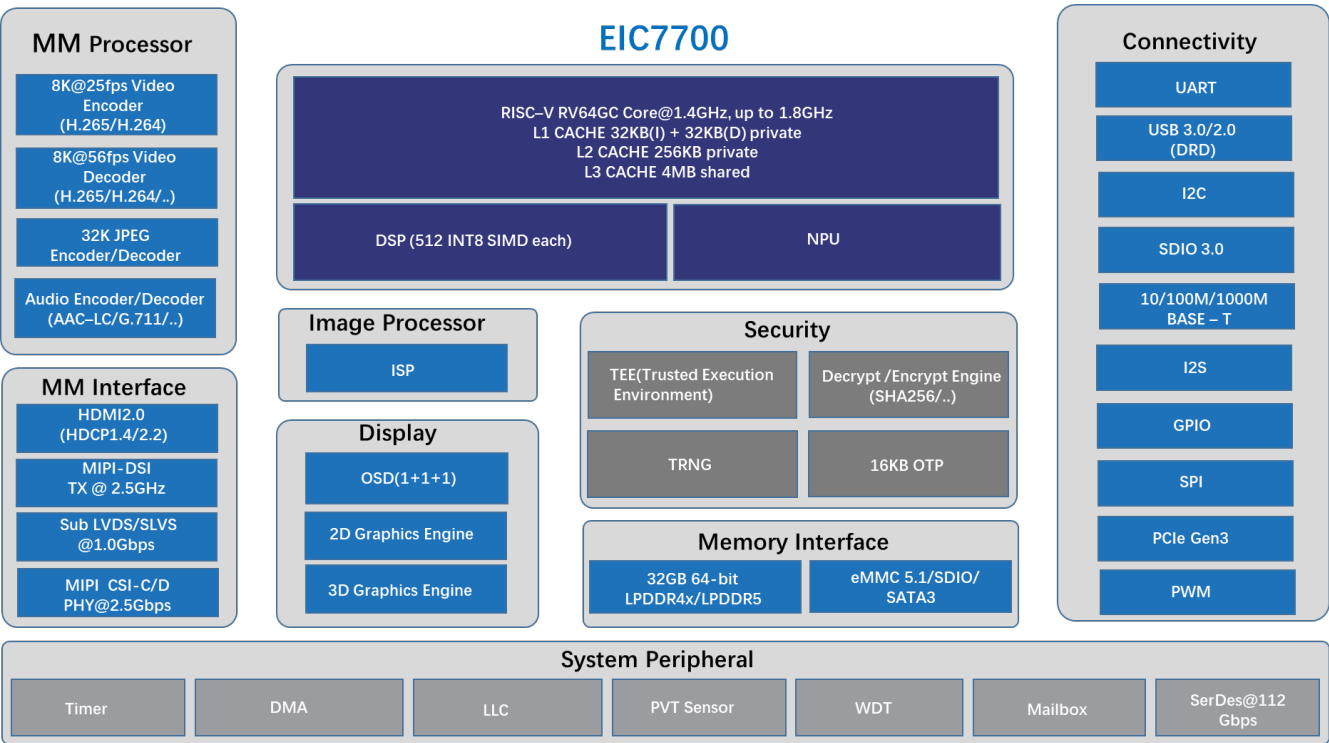


Figure 2: EIC7700 Chip Block Diagram

3.2. System Block Diagram

3.2.1. P550S System Block Diagram

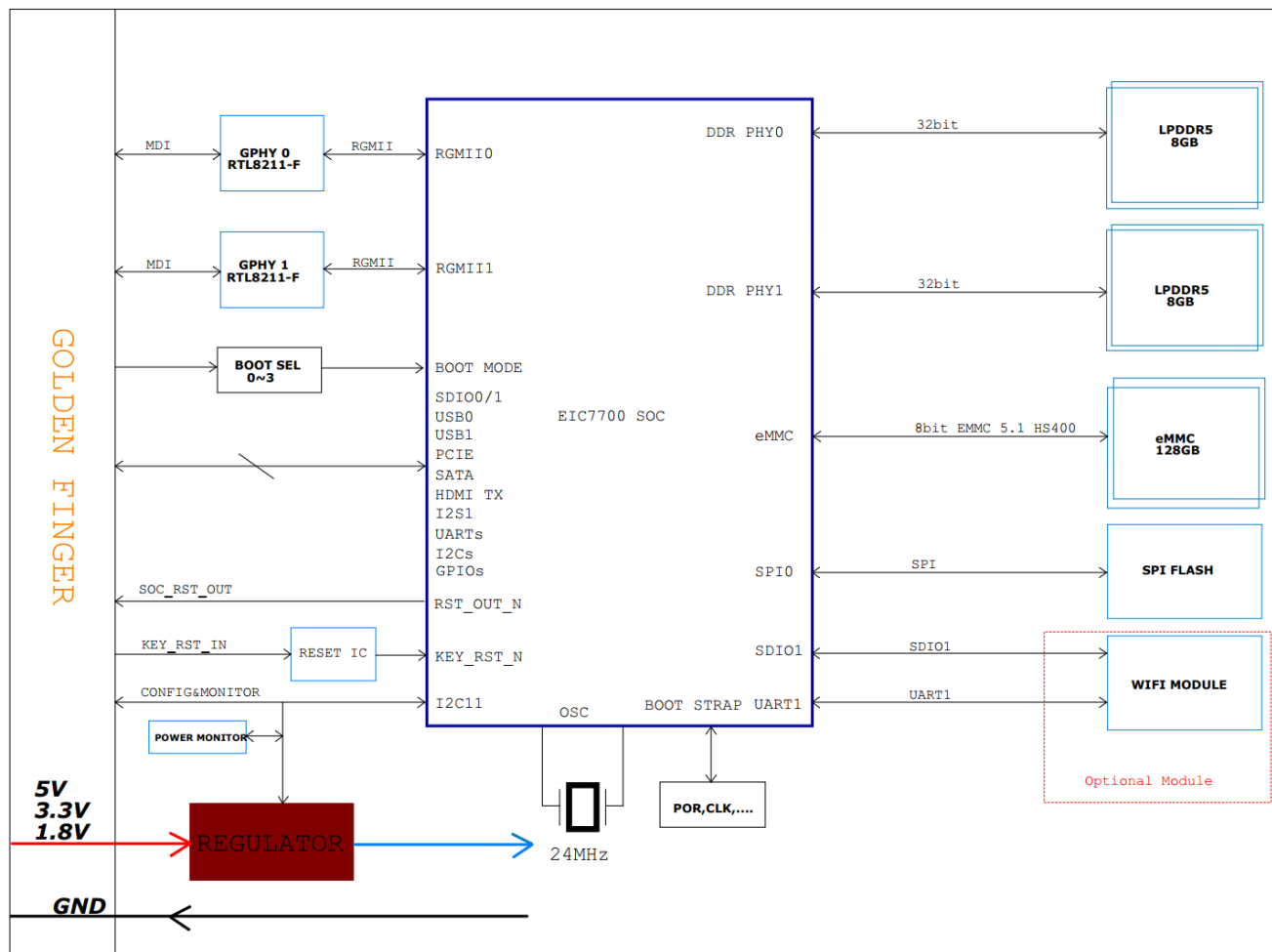


Figure 3: P550S System Block Diagram

3.2.2. P550C System Block Diagram

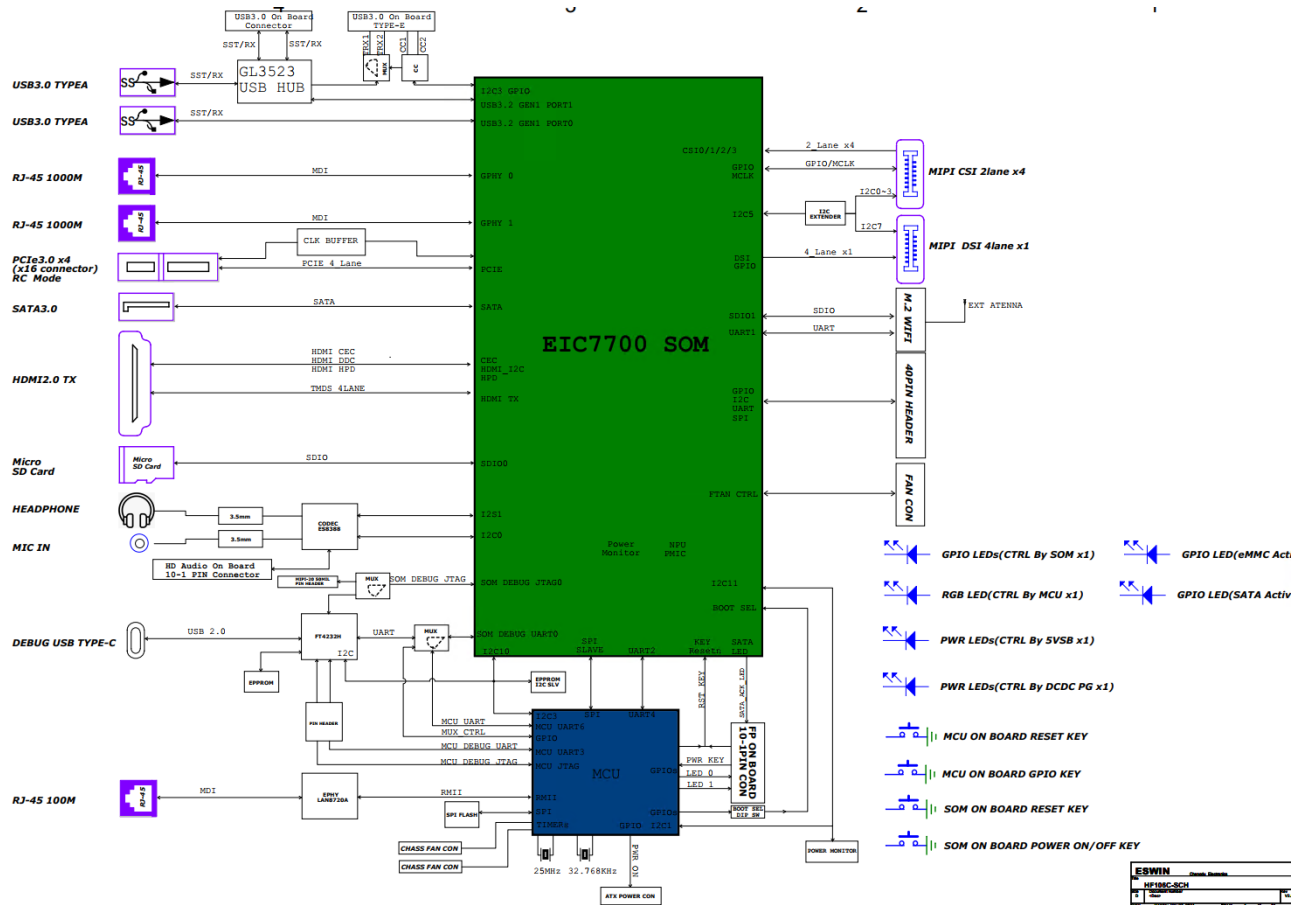


Figure 4: P550C System Block Diagram

3.3. Power Supply Block Diagram

3.3.1. P550S Power Supply Block Diagram

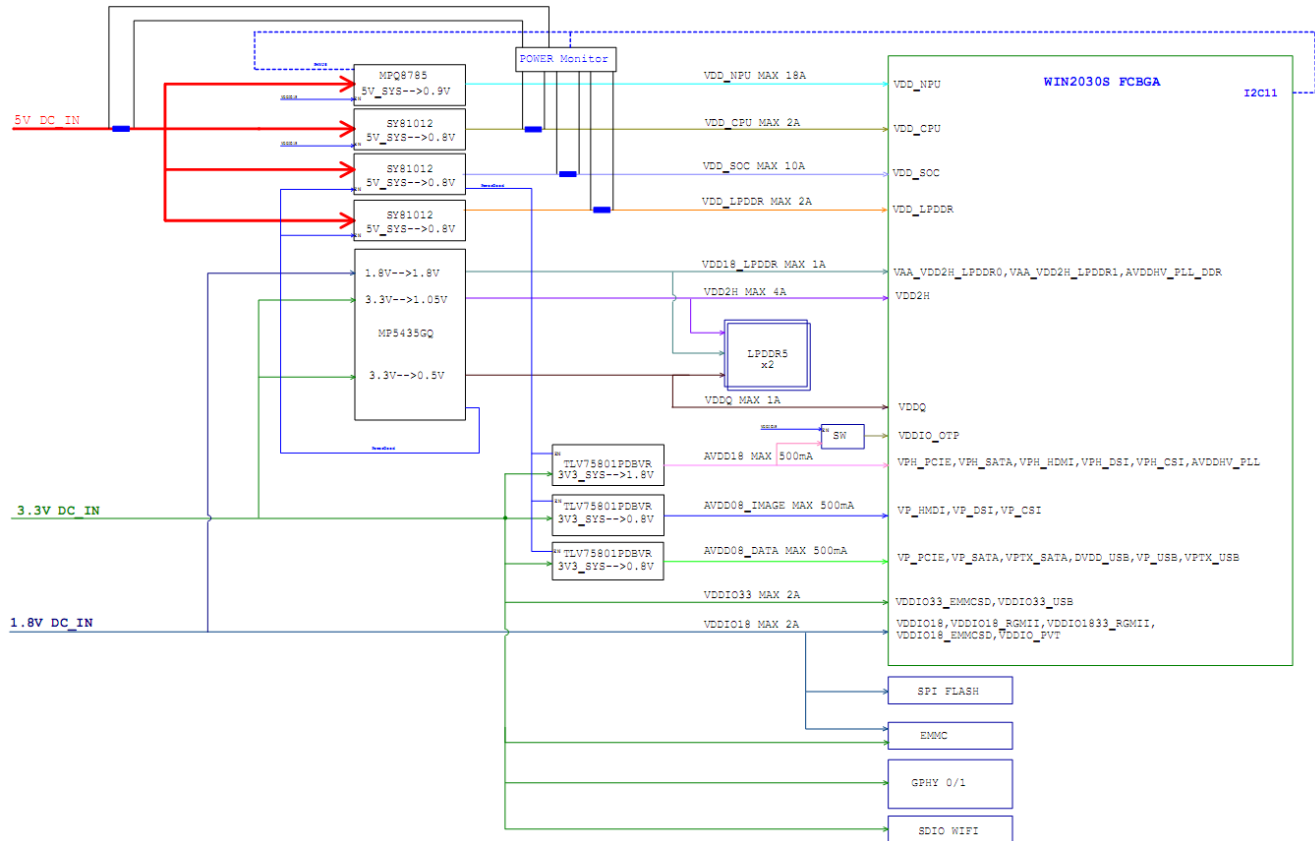


Figure 5: P550S Power Supply Block Diagram

3.3.2. P550C Power Supply Block Diagram

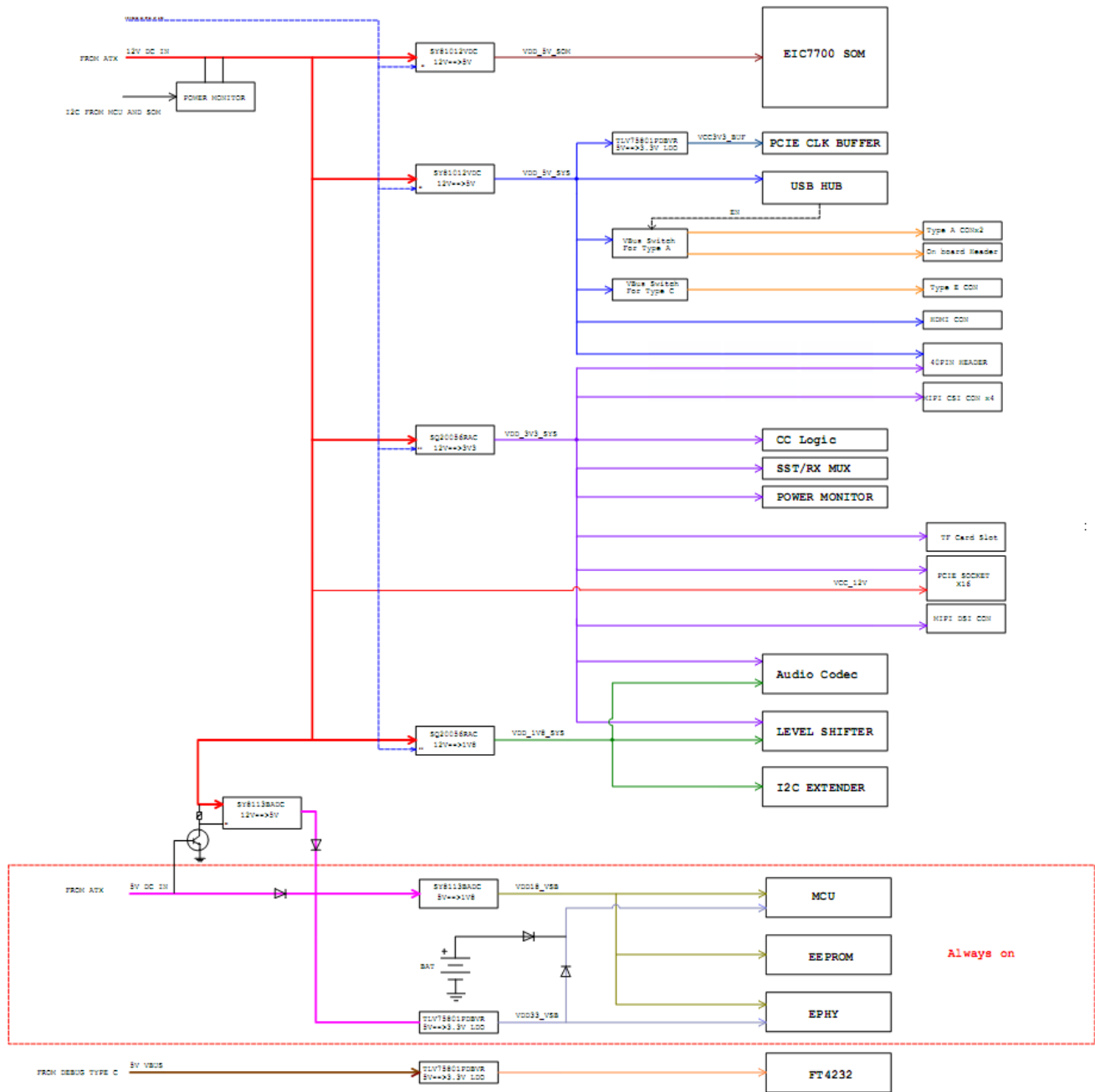


Figure 6: P550C Power Supply Block Diagram

4. Hardware Introduction and Usage Guide

4.1. Preparation

Before powering on the development board, please complete all the preparations in this section to ensure the device is safe and operates correctly.

4.1.1. Heatsink and Fan

To prevent the core chip from overheating and sustaining damage, you must install the heatsink and connect the fan power before the first power-on.

Warning

Powering on without installing the heatsink or connecting the fan may cause permanent damage to the EIC7700 SoC due to overheating.

Steps:

1. Align the heatsink with the SoC chip and secure it firmly using screws or clips.
2. Locate the on-board cooling fan power connector (usually marked as FAN) and plug the fan's power cable into it.

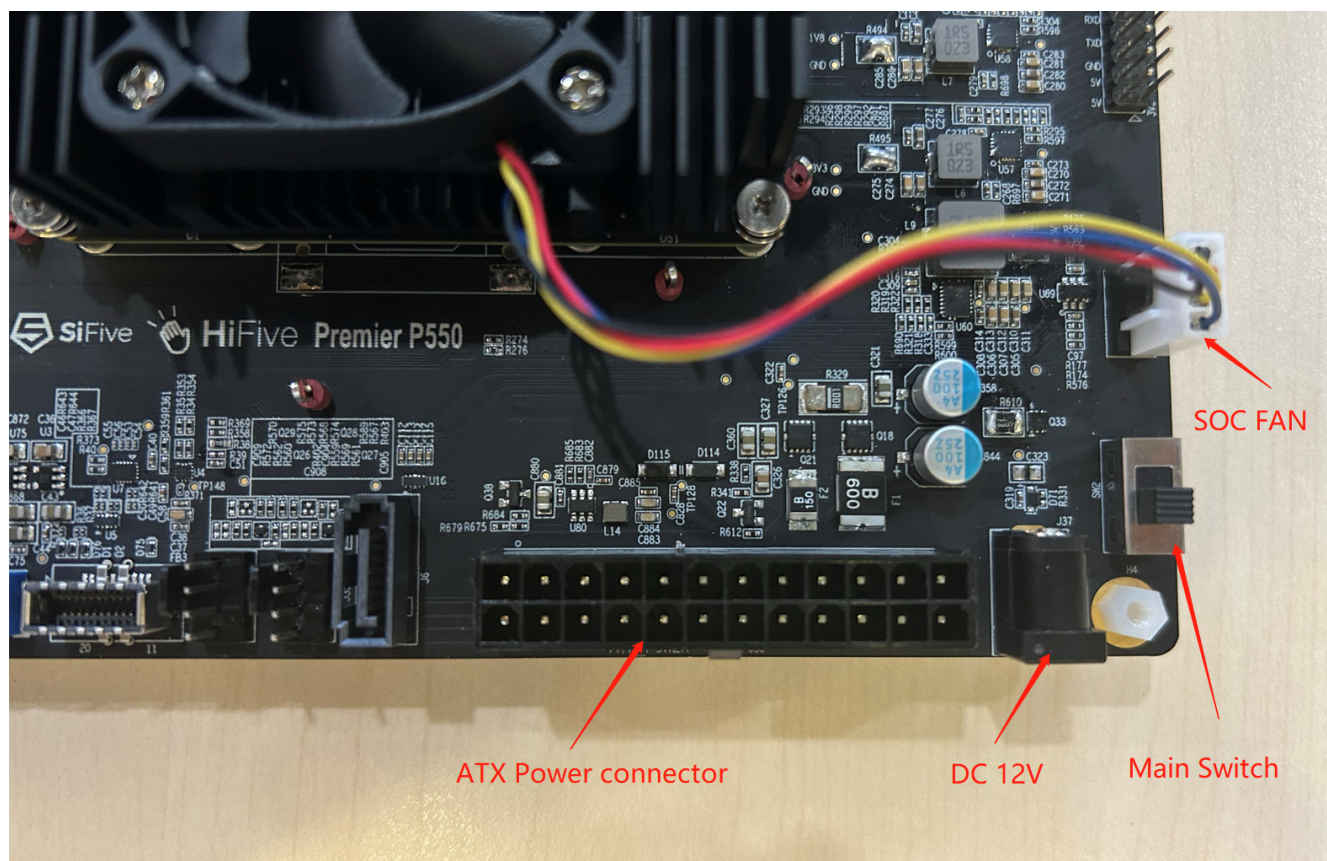


Figure 7: Power and Fan Interface Layout

4.1.2. Connecting Peripherals

For convenient debugging, operation, and feature verification, it is recommended to connect the following common peripherals before the first power-on. Refer to [Interface Details](#) for port locations.

Peripheral	Connection Interface	Main Purpose
Debugging & Command Line	Rear Panel USB Type-C Port	Core interface for debugging and flashing: Connects to the host PC. Before first use, the FT4232H driver must be installed on the PC. After installation, this port will be recognized as a virtual serial port (COMx/ttyUSBx), serving as the primary channel for system image flashing, serial debugging, and console login. For driver installation, refer to Driver Installation .
Display Output	Rear Panel HDMI Port	Connects to a monitor to view the graphical desktop or boot interface.
Input Devices	Rear Panel USB Type-A Port	Connects keyboard and mouse for operating the GUI or command line after boot.
Network	Rear Panel Gigabit Ethernet (RJ45) Port	Connects a network cable to access the LAN, facilitating network installation, updates, and remote access.
Others	Audio ports, etc.	Connect speakers, microphones, and other peripherals as needed.

4.1.3. Powering On

The P550 development board supports the following two power supply methods; choose either one:

- 24-Pin ATX Power Supply: Uses a standard desktop power supply.
- DC 12V Power Adapter: Uses a 12V/5A DC power supply.

Steps:

1. Connect the selected power cable (ATX or DC adapter) to the corresponding interface on the development board.
2. Locate the main power switch (Main Switch) on the board and toggle it to the ON position. The LED D110 on the P550C will light up, and the RGB light will turn red, indicating that the board is powered and in a standby state. For LED locations and definitions, refer to [Status LEDs](#) in this document.

Tip

The Main Switch acts as the main power cutoff. Turning it off (OFF) completely cuts power to the board, suitable for long-term storage or hardware modification.

The pin definition for the ATX power interface is shown below for users who need to customize cables:

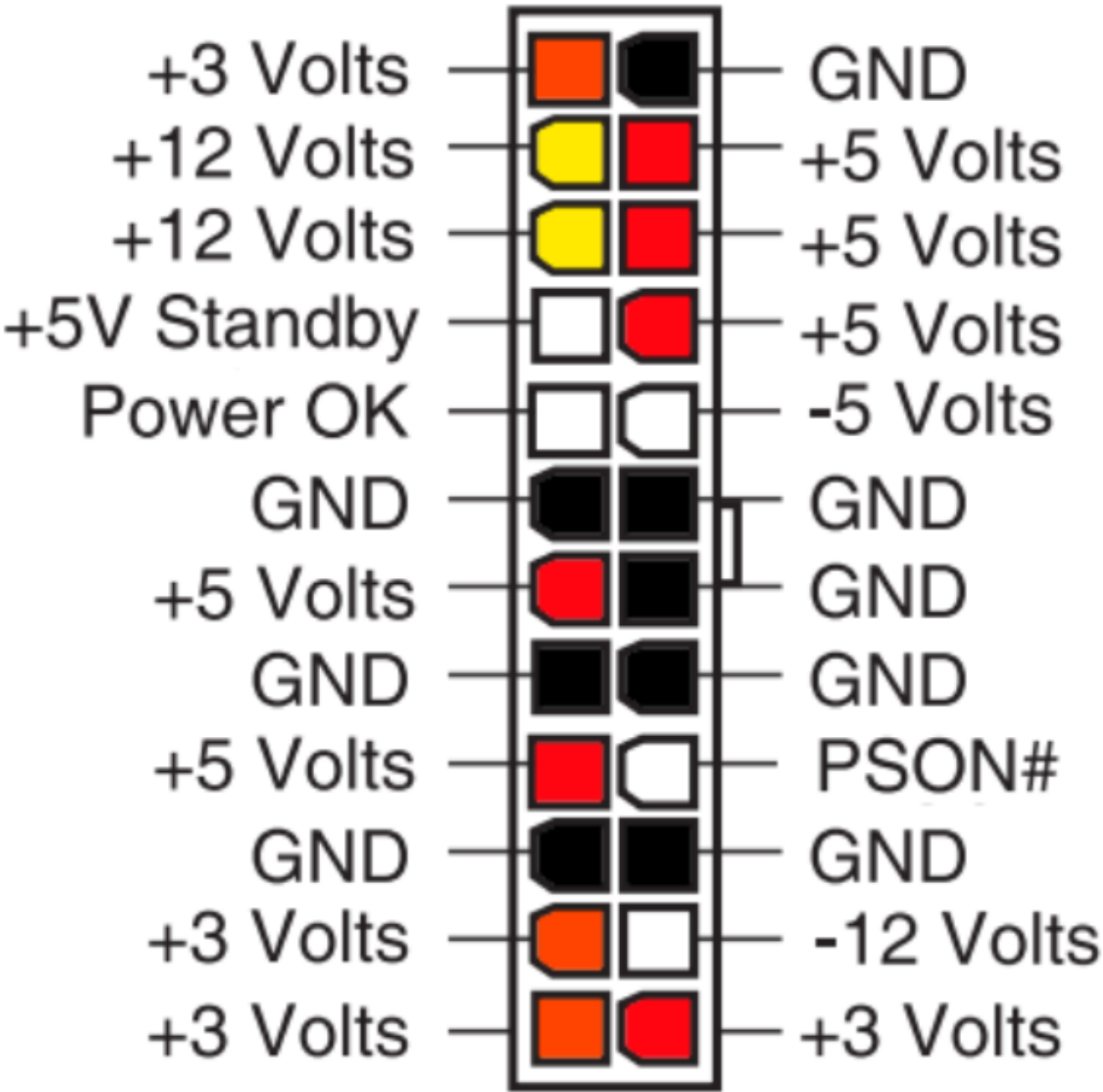


Figure 8: ATX Power

4.2. Power On/Off and Operation

After completing the hardware preparation, you can use this section to learn how to power on/off the development board, select boot modes, and identify system status through buttons and LEDs.

4.2.1. Button Functions and Operation

The P550 development board provides four physical buttons for controlling system power and reset. Their locations are shown below:

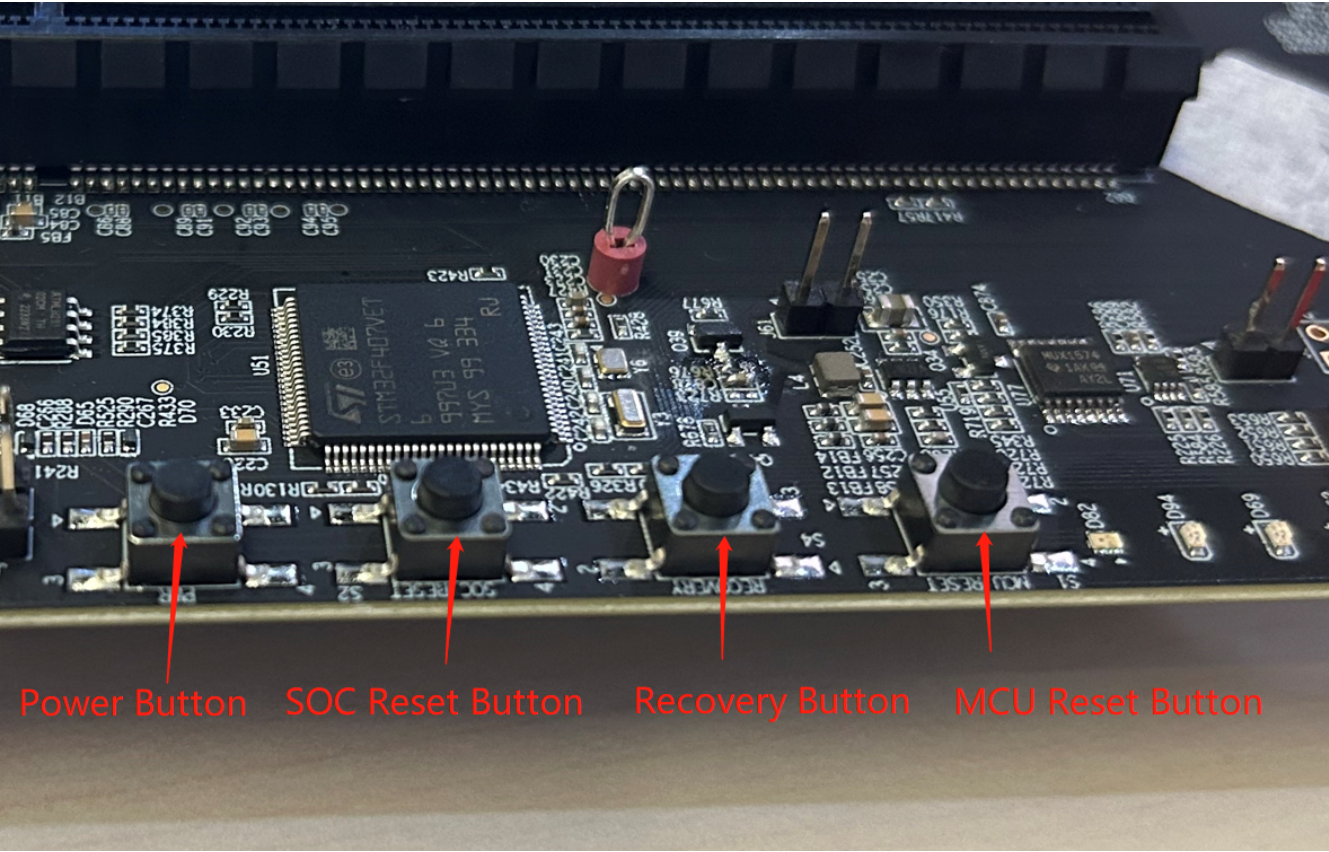


Figure 9: Buttons

The functions and operation methods for each button are as follows:

Button Name	Main Function	Operation Method
Power Button	Controls system power-on and forced shutdown	Short press (approx. 1 sec): Used to power on when the system is off.Long press (approx. 10 sec): Used to force power-off when the system is unresponsive.
SOC Reset Button	Resets the main system (EIC7700 SoC)	Short press: Restarts the main system (EIC7700 SoC), similar to a PC “Reset” button.
MCU Reset Button	Resets the MCU	Short press: Restarts the MCU. The P550 includes an MCU for remote management.
Recovery Button	Reserved Function	This button function is not currently enabled in this version.

Tip

1. **Normal shutdown** should be performed by executing a shutdown command in the operating system (e.g., typing `poweroff` in the Linux terminal). The long-press operation of the **Power Button** is a **forced power-off**, to be used only when the system hangs, and may cause data loss.
2. While the system is running, do not press the **SOC Reset Button** and **MCU Reset Button** arbitrarily.

4.2.2. Status LEDs

The on-board LED indicators are the most intuitive tool for diagnosing system operating status. The table below details the meaning of each LED:

Table 3: P550 Status Indicator Description

LED Name	Component Designator	Status and Meaning
MCU System Power LED	D110	Solid Green: Indicates the Management Unit (MCU) is powered correctly, and the development board is connected to power (Main Switch is ON).
SOC System Power LED	D74	Solid Green: Indicates the main system (SoC) is powered correctly, usually lights up after you press the Power Button to turn on.
RGB Status LED	D82	System core status indicator: • Red: MCU is powered, but SoC is not started (Power connected, not booted). • Green: SoC is in the boot process (U-Boot, Kernel loading). • Blue: SoC boot complete, Operating System is running normally. • Other colors: Can be customized by user programs to indicate specific application states.
User Defined LED	D94	Can be controlled by user via GPIO programming for project-specific indications.
eMMC Activity LED	D69	Flashes when the eMMC storage chip is performing read/write operations. Can also be configured as a standard GPIO.
SATA Activity LED	D62	Flashes when a device connected to the SATA interface is performing read/write operations. Can also be configured as a standard GPIO.

4.2.3. Power On/Off Operation Procedure

Combining buttons and LEDs, the standard operation procedure is as follows:

1. Power On:

- Ensure power is connected and the **Main Switch** is turned on. At this point, D110 (MCU LED) and D82 (RGB LED) should be red.
- Short press the **Power Button**.
- Status change: The fan starts spinning, D74 (SOC LED) lights up green, D82 (RGB LED) turns green (blinking or solid, booting), and finally changes to blue, indicating that system boot is complete.

2. Normal Shutdown (Recommended Method):

- In the running Operating System (e.g., Linux), execute the shutdown command.
- Status change: After the system finishes its processes, D74 (SOC LED) turns off, D82 (RGB LED) turns back to red, and the fan stops.

3. Forced Shutdown (Emergency Method):

When the operating system is unresponsive, long press the **Power Button** for about 10 seconds until D74 (SOC LED) turns off and D82 (RGB LED) turns back to red.

Warning

This operation cuts power directly and may cause loss of unsaved data.

4.2.4. Boot Mode Selection

The P550 uses a 4-position DIP switch (SW1) to select the Bootloader source, mainly used for system repair and upgrades.

- 1. **Switch Description:** As indicated by the silk screen in the figure, toggling towards the side marked ON represents 0, and toggling towards the side marked with numbers 1, 2, 3, 4 represents 1.
- 2. **Mode Selection:** Below are the two most commonly used settings. Please toggle according to the figure:

DIP Switch State (SW1[0:3])	Boot Mode	Applicable Scenario
0100	Boot from On-board SPI Flash	Regular usage scenario, booting the installed OS from the board's own storage.
1100	Boot from USB (Device Mode)	Repairing bootloader. When the system in SPI Flash is corrupted, the board simulates a USB drive to allow copying a new Bootloader from the host for recovery. For detailed steps, refer to Scenario 2: Bootloader Unable to Start (USB Mode Recovery) .

- Set DIP switch to 0100 to boot from SPI Flash .

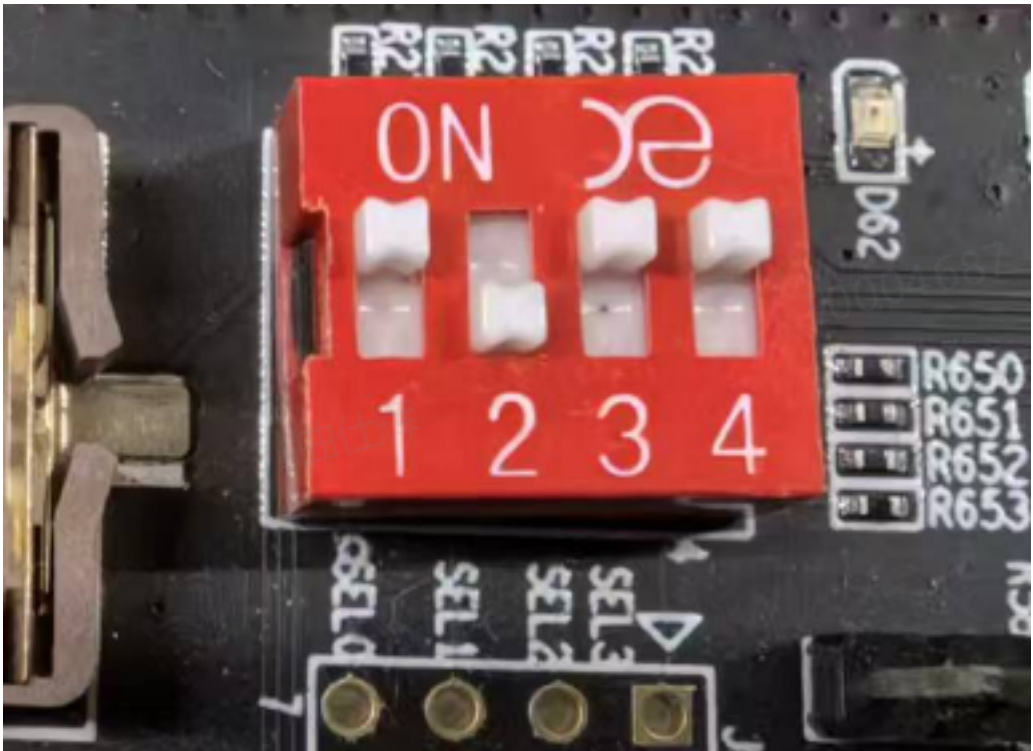


Figure 10: SPI Flash Boot

- Set DIP switch to 1100 to boot from USB .

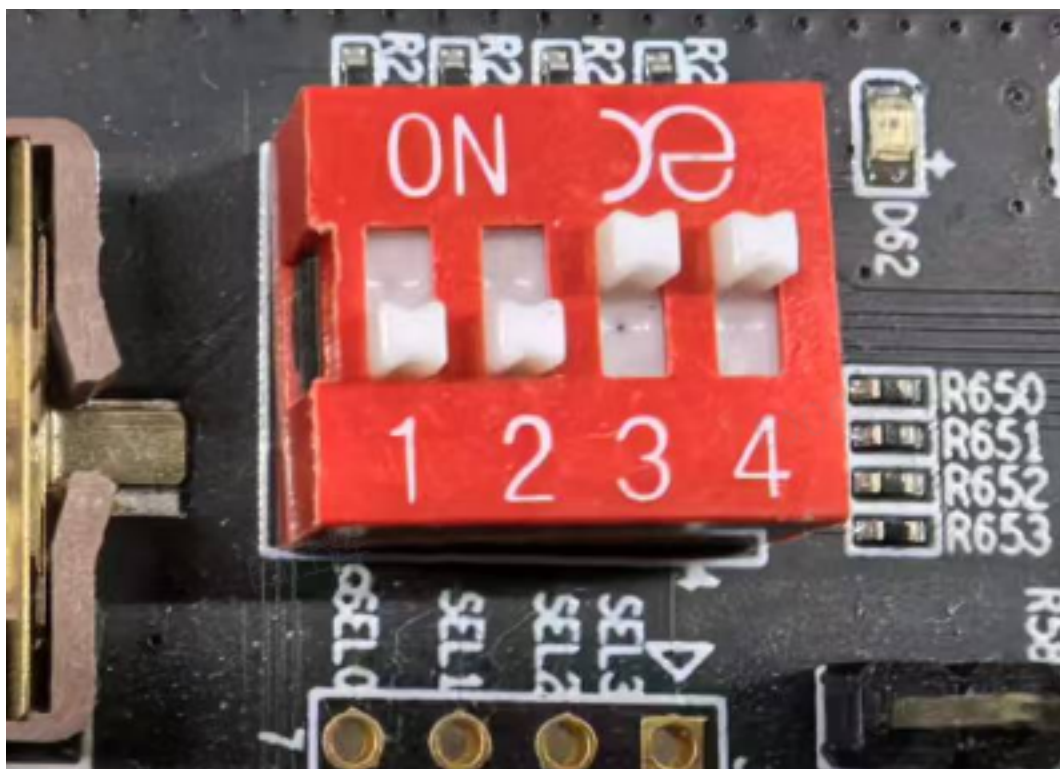


Figure 11: USB Boot

Tip

Changes to DIP switch settings must be made after the development board is completely powered off (turn off **Main Switch** or unplug power). After setting, power on again for the new settings to take effect.

4.3. Interface Details

The P550 development board provides a rich set of external interfaces for connecting peripherals, expanding functionality, and system debugging. This chapter details the location, function, and typical usage of all interfaces.

4.3.1. Rear Panel Interfaces

The rear panel integrates the most commonly used input/output interfaces, allowing users to quickly connect standard peripherals.

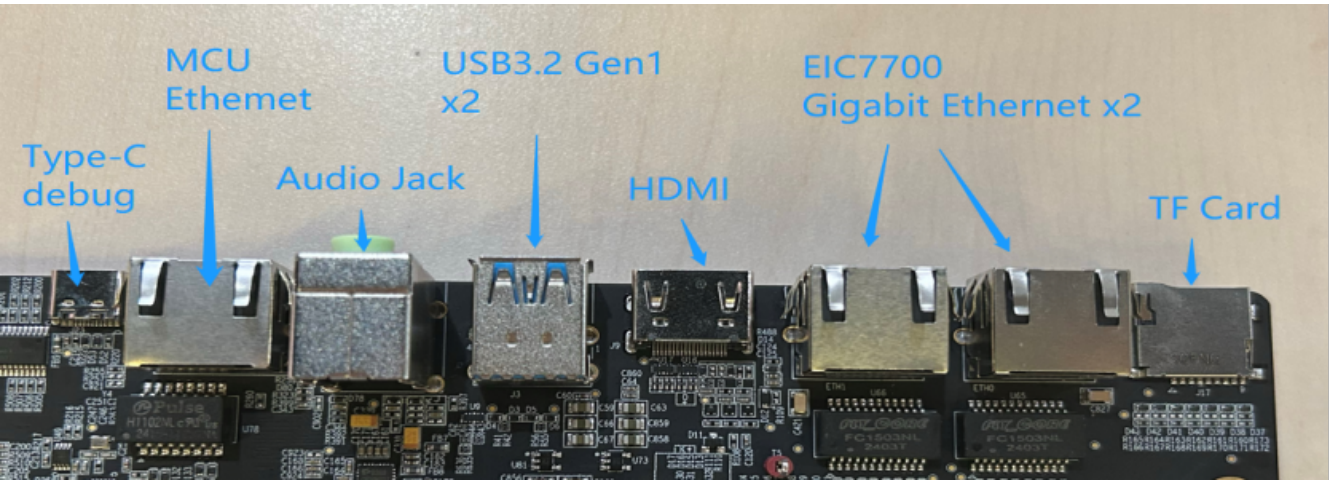
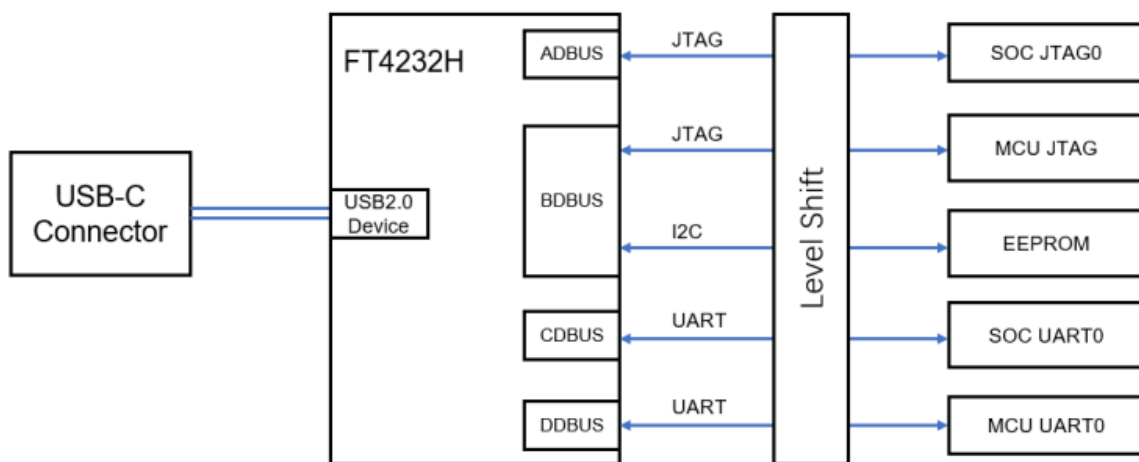


Figure 12: Rear Panel Interfaces

The table below details the function of each interface:

Interface	Quantity	Function Description and Usage
HDMI Out	1	Video Output: Connects to a monitor, supporting up to 4K resolution. Used for displaying the graphical desktop or system boot interface.
Gigabit Ethernet	2	Wired Network: Connects to a router or switch to provide high-speed network access for the development board. The two ports can be used to isolate different networks.
Audio I/O	1	Audio: The green port is for audio output (headphones/speakers); the pink port is for microphone input.
USB 3.2 Gen1 Type-A	2	Universal Peripherals: Connects keyboards, mice, USB drives, external hard drives, and other USB devices. The upper one is the USB0 port (configurable as a programming port), and the lower one is the USB1 port.
TF Card Slot	1	Storage Expansion: Inserts a Micro SD (TF) card, which can serve as additional storage or a system boot medium.
USB Type-C (Debug)	1	Core Debugging and Flashing Interface: Connects to the host PC. This interface converts to JTAG and UART signals via the on-board FT4232H chip, serving as the primary channel for serial debugging, and console login. For driver installation, refer to Chapter 5 Windows Driver Installation and Debugging Guide .
MCU Ethernet (Management)	1	Management Port: Connects to the MCU's 100Mbps Ethernet port, enabling out-of-band management features like remote power management and status monitoring, usable even when the main system (SoC) is off.

FT4232H Chip Block Diagram: This chip expands the single USB Type-C interface into multiple debugging signals. Its internal architecture is shown below.



4.3.2. Front Panel Interfaces

The P550 supports standard Mini-DTX chassis. Front panel cables (such as audio, USB, switches) from the chassis must be connected to the corresponding sockets on the board.

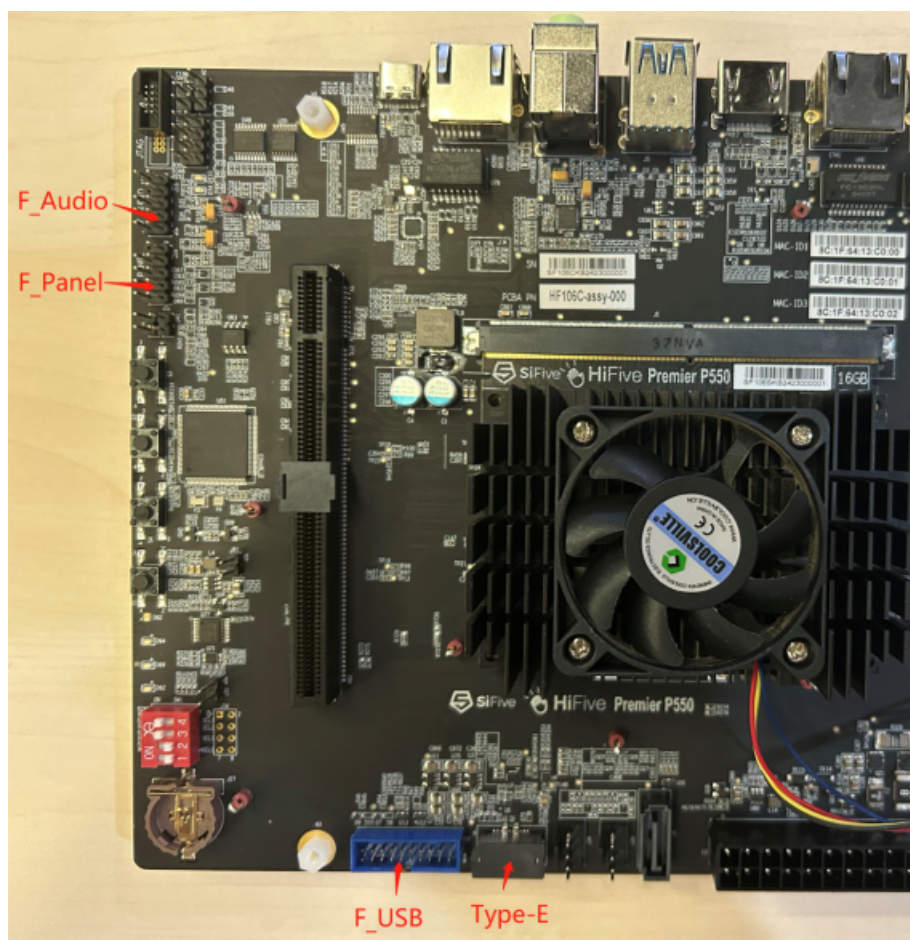


Figure 13: Front Panel Socket Locations

The definitions of each socket and corresponding cable plugs are shown below. When connecting, ensure the plug orientation aligns with the socket keying.

• **Front Panel Audio Socket (F_Audio, J16):**

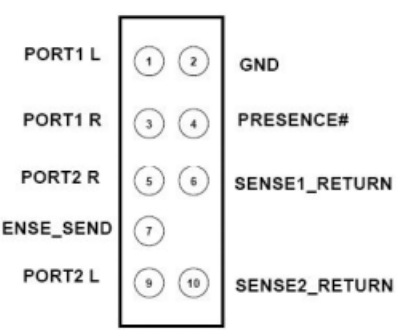


Figure 14: Front Panel Audio Socket Pin Definition

• **Front Panel Control Socket (F_Panel, J28):** Used to connect chassis power switches, reset switches, and hard drive LEDs.

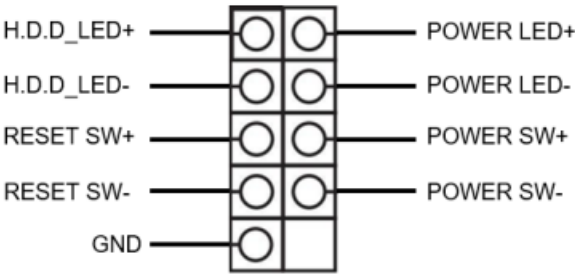


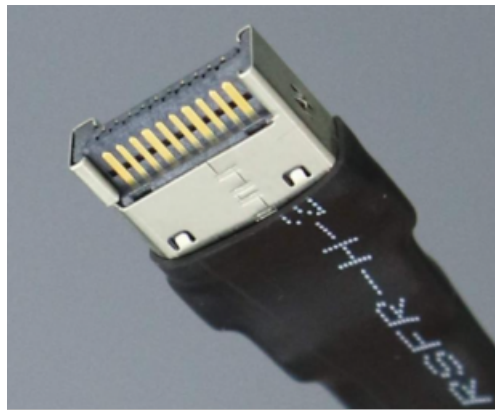
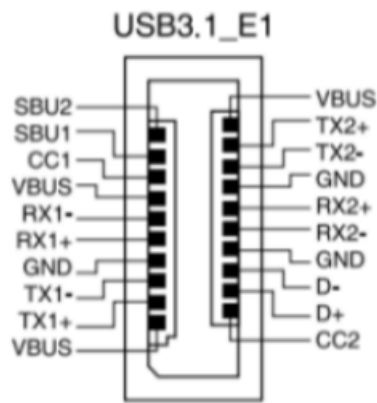
Figure 15: Front Panel Control Socket Pin Definition

• **Front Panel USB Socket (F_USB, J4):** Used to expand USB Type-A interfaces on the front of the chassis.

Pin No.	Definition	Pin No.	Definition
1	VBUS	11	D2+
2	SSRX1-	12	D2-
3	SSRX1+	13	GND
4	GND	14	SSTX2+
5	SSTX-	15	SSTX2-
6	SSTX+	16	GND
7	GND	17	SSRX2+
8	D1-	18	SSRX2-
9	D1+	19	VBUS
10	NC		



• **Type-E Connector (J2):** Used to expand USB Type-C interfaces on the front of the chassis. Chassis typically provide Type-E to Type-C cables.



4.3.3. Wireless Expansion Interface

The development board reserves an M.2 E-Key slot for installing wireless network modules supporting the SDIO protocol to add Wi-Fi and Bluetooth functionality.

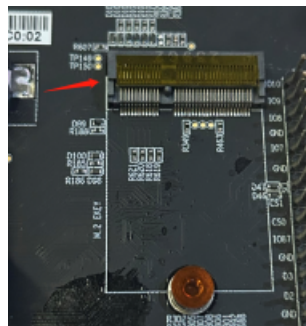


Figure 16: SDIO WiFi Module M.2 E-Key Interface

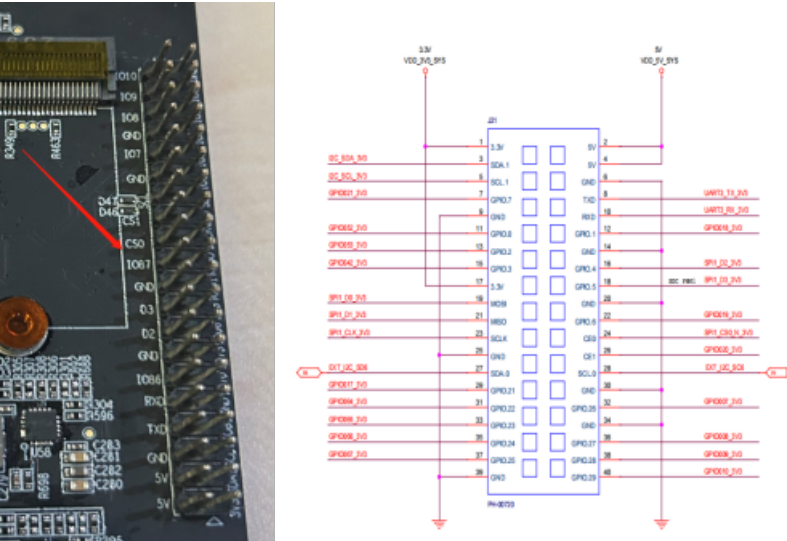
Tip

- Please ensure the selected wireless module has an M.2 E-Key interface and supports the SDIO protocol. ESWIN has released a wireless module; contact ESWIN sales for purchase.
- Ensure the development board is completely powered off before installation.

4.3.4. Expansion and Debugging Interfaces

4.3.4.1. 40-Pin Expansion Header

This header exposes unused system signals (such as GPIO, I2C, SPI, PWM, etc.) for advanced users to perform custom hardware functionality expansion.



4.3.4.2. Test Headers

The board provides multiple sets of test Headers, primarily intended for advanced debugging, system recovery, and functional expansion. **Most test Header are intended for development or advanced use only and do not require action by general users.**

Warning

Before operating any test headers, ensure the development board is completely powered off.

The table below details the function, typical usage scenarios, and operation guidelines for each test header:

Table 4: Test Headers

Pin	Function Description
J53	SOC JTAG0 Selection: 1. Without jumper: SOC JTAG0 connects to FT4232H 2. With jumper: SOC JTAG0 connects to MIPI20 (J59)
J42	Test UART0: 1. UART0_TX 2. UART0_RX 3. GND
J24	Test MCU JTAG and Uart
J41	MCU Boot Mode Selection, typically unused: 1. Without jumper: MCU boots from main Flash 2. With jumper: MCU boots from embedded SRAM
J55	Test SOC Reset Signal: 1. KEY_RESET_N 2. GND
J61	When the system is powered by a 12V DC adapter and the MCU is not working, use this jumper to power the SOC system

Table 4: Test Headers

Pin	Function Description
J31	When the system is powered by an ATX power supply and the MCU is not working, use this jumper to output 12V power

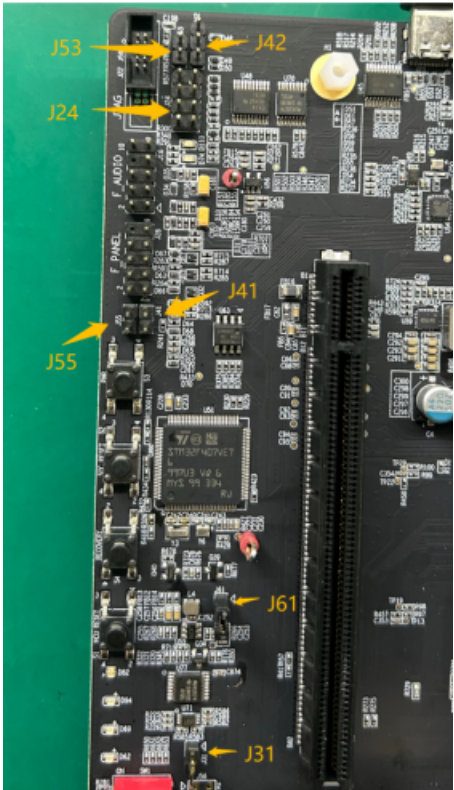


Figure 17: Test Headers

4.3.4.3. GND Test Rings

T1, T2, T3, T4, T5, T6, T7, T8 are GND test rings, convenient for grounding during measurements.

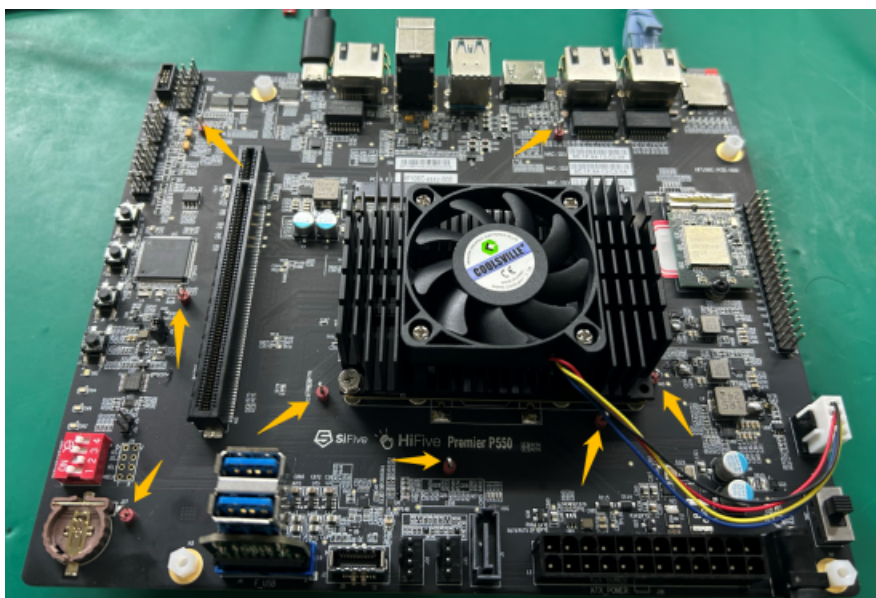


Figure 18: GND Test Rings

5. Windows Driver Installation and Debugging Guide

Before using the P550 development board for serial debugging, console login, JTAG debugging, or Fastboot flashing under the Windows operating system, you must install the corresponding USB drivers. Linux users do not need to install additional drivers.

This chapter will guide you through the installation of the following core drivers:

1. **FT4232H Driver:** Used to enable Serial (COM) and JTAG functions for the Type-C debug interface.
2. **Fastboot Driver:** Used for fast flashing of system images via USB.

5.1. FT4232H Driver Installation (Serial & JTAG)

The USB Type-C debug interface on the P550 rear panel uses the onboard FT4232H chip to convert a single USB connection into four independent virtual devices: two JTAG channels and two UART channels, corresponding to the EIC7700 SoC and MCU respectively.

5.1.1. Obtaining the Driver Installation Tool

1. [Click to Download](#) the driver installation tool: `UsbDriverTool-2.1.exe`.
2. Save it locally (e.g., `D:\`), run it, and extract it to the default `UsbDriverTool` directory.

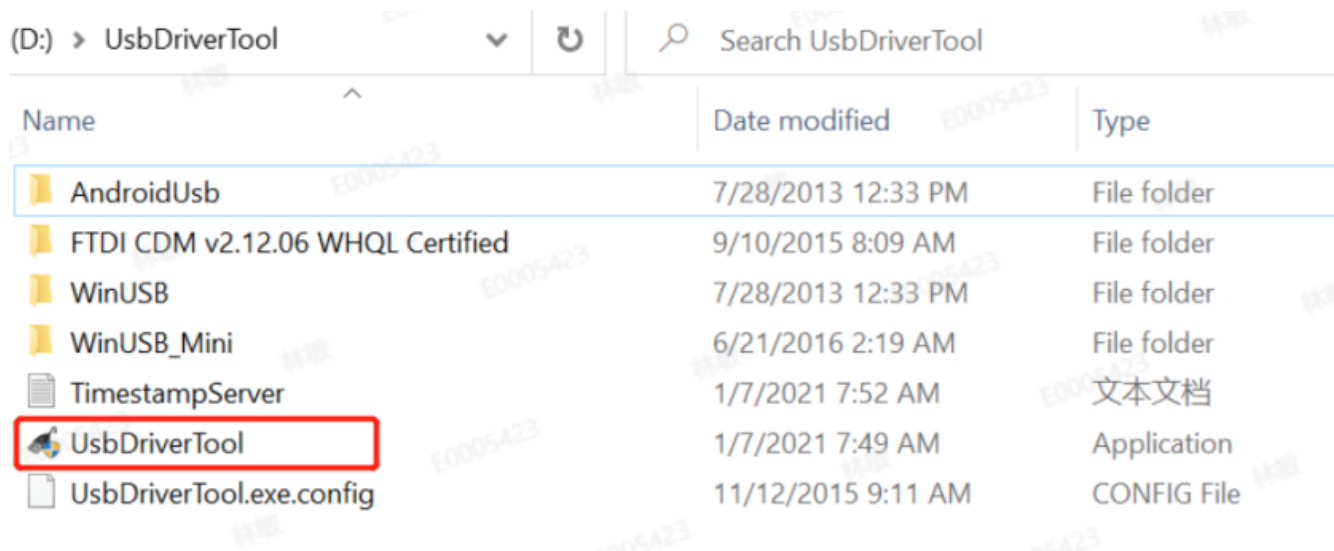


Figure 19: UsbDriverTool

5.1.2. Connect Device and Launch Tool

1. Use the USB Type-C data cable from the accessories to connect the P550's rear panel Type-C interface to your Windows PC.
2. Open the `UsbDriverTool` directory extracted in the previous step, and double-click to run `UsbDriverTool.exe`.

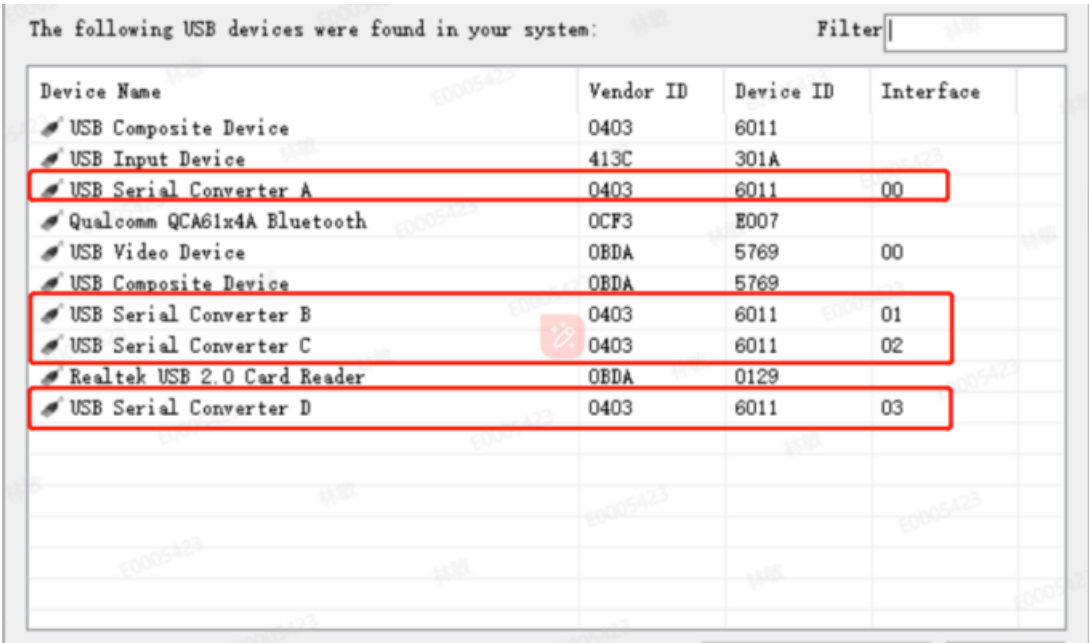


Figure 20: USB Devices

The program will automatically identify four uninstalled devices with DeviceID 6011 , which correspond to:

Device Name	Vendor ID	Device ID	Interface	Description
USB Serial Converter A	0403	6011	00	EIC7700 SoC JTAG
USB Serial Converter B	0403	6011	01	MCU JTAG
USB Serial Converter C	0403	6011	02	EIC7700 SoC UART (Serial Port)
USB Serial Converter D	0403	6011	03	MCU UART (Serial Port)

5.1.3. Installing the Serial (UART) Driver

The serial driver creates virtual COM ports on the PC, allowing access to the development board’ s console via serial tools (e.g., Putty, MobaXterm).

- 1. In the UsbDriverTool interface, use the left mouse button to select the following two devices:
 - 02 - USB device (EIC7700 SoC UART)
 - 03 - USB device (MCU UART)
- 2. Right-click on the selected devices and choose Install FTDI CDM from the context menu.

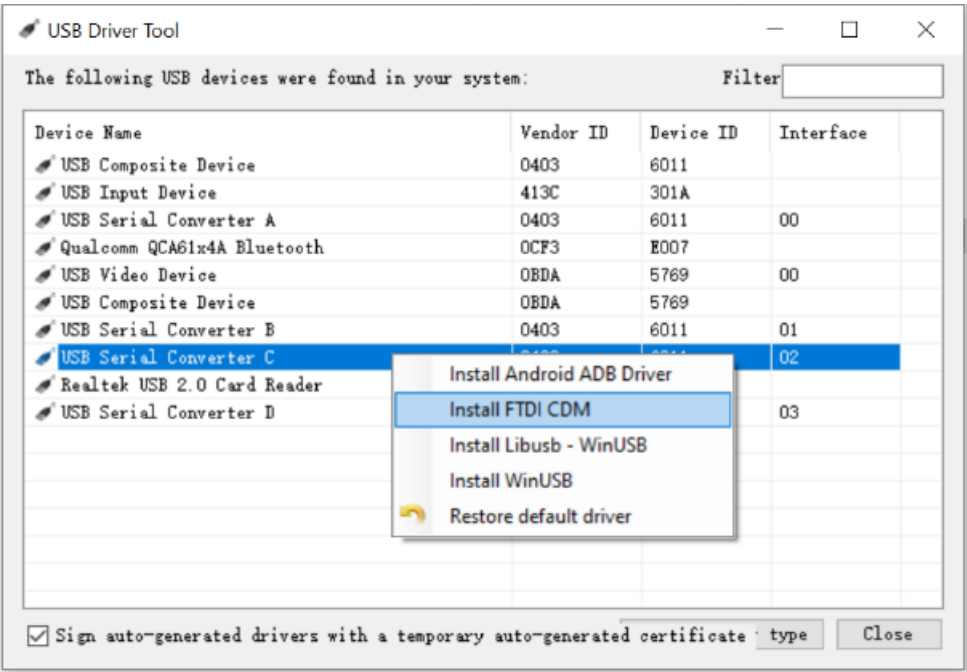


Figure 21: Install UART Driver

3. Installation Verification: Open Windows **Device Manager** and expand the **Ports (COM & LPT)** list. You should see four new serial ports, for example, **COM3 - COM6**.
- **Normally**, the third port in the list (e.g., **COM5**) corresponds to the EIC7700 SoC UART.
 - **Normally**, the fourth port in the list (e.g., **COM6**) corresponds to the MCU UART.

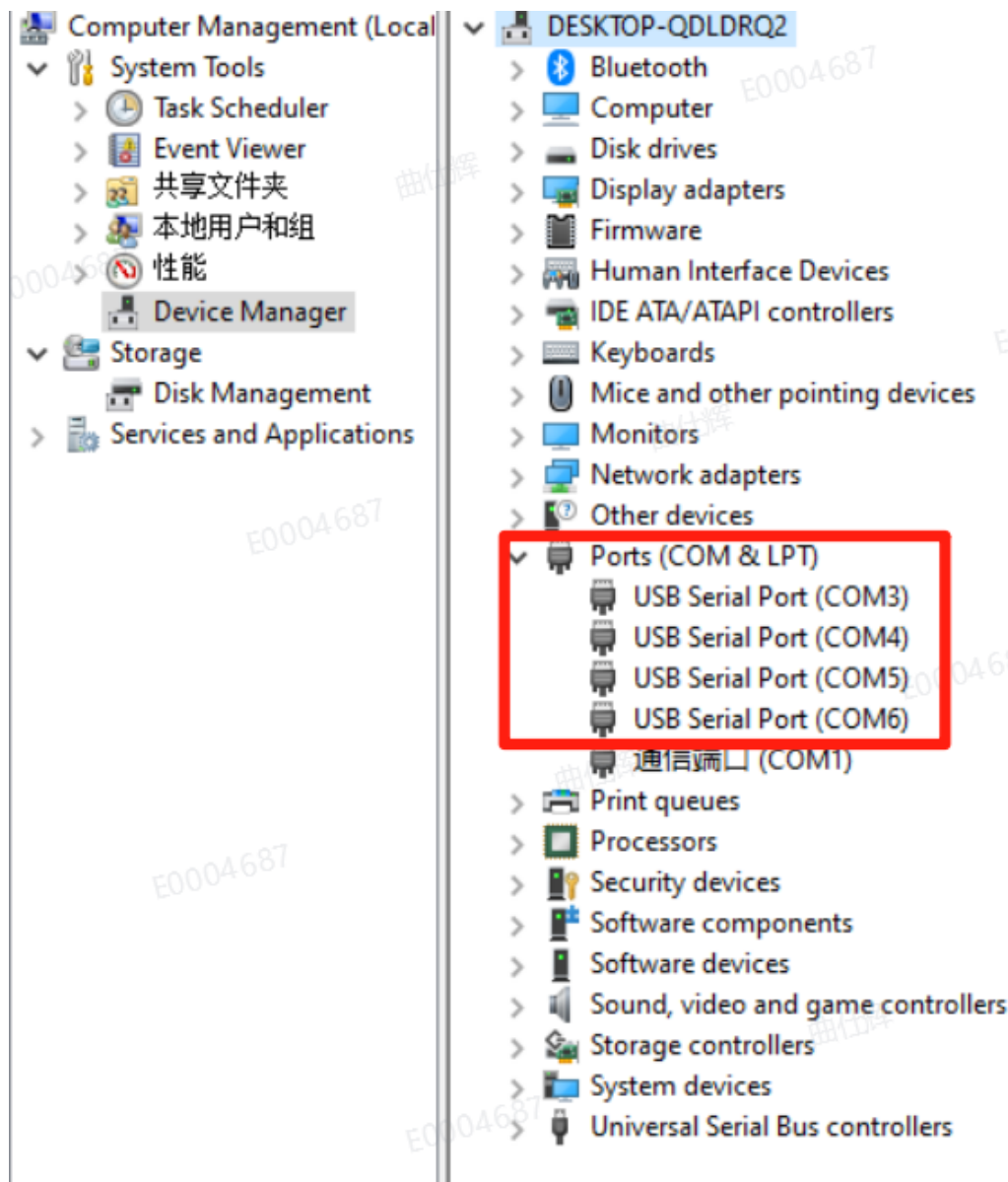


Figure 22: Serial Devices

Tip

The specific COM port numbers (e.g., COM5, COM6) vary by computer; please refer to the actual display in Device Manager.

Performing Serial Debugging:

Once the driver is successfully installed, you can access the development board using a serial terminal tool:

1. Open your preferred serial terminal software (e.g., Putty).
2. Select the corresponding COM port (for the main system, usually select the third port, e.g., COM5).
3. Set the baud rate to 115200 8N1.
4. After connecting, power on the development board, and you will see boot messages in the terminal and can interact via the command line.

5.1.4. Installing the JTAG Driver

The JTAG driver supports low-level chip debugging via the USB interface.

- 1. In the UsbDriverTool interface, use the left mouse button to select the following two devices:
 - 00 - USB device (EIC7700 SoC JTAG)
 - 01 - USB device (MCU JTAG)
- 2. Right-click on the selected devices and choose Install WinUSB from the context menu.

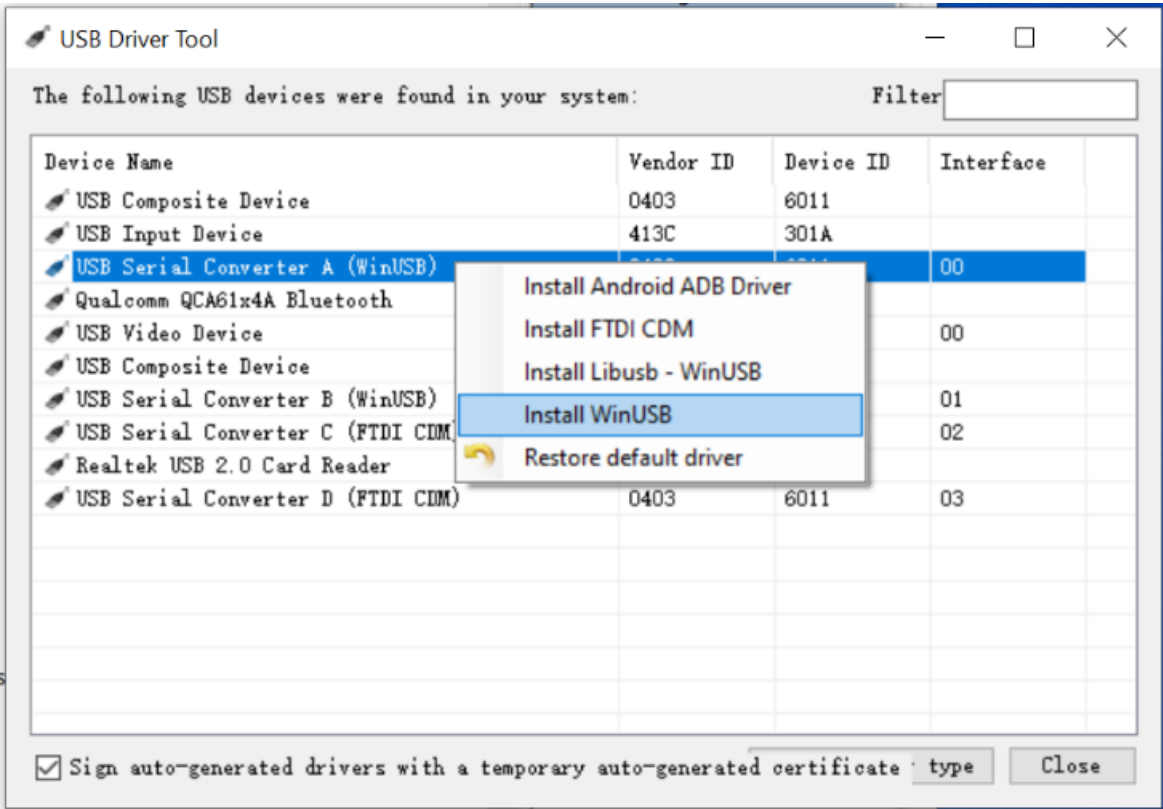


Figure 23: Install JTAG Driver

- 3. Installation Verification: Open “Device Manager” again and expand the “Universal Serial Bus controllers” list. You should see two devices named Quad RS232-HS . At the same time, the “Ports (COM & LPT)” list should only retain the two serial port devices installed previously (e.g., COM5, COM6).

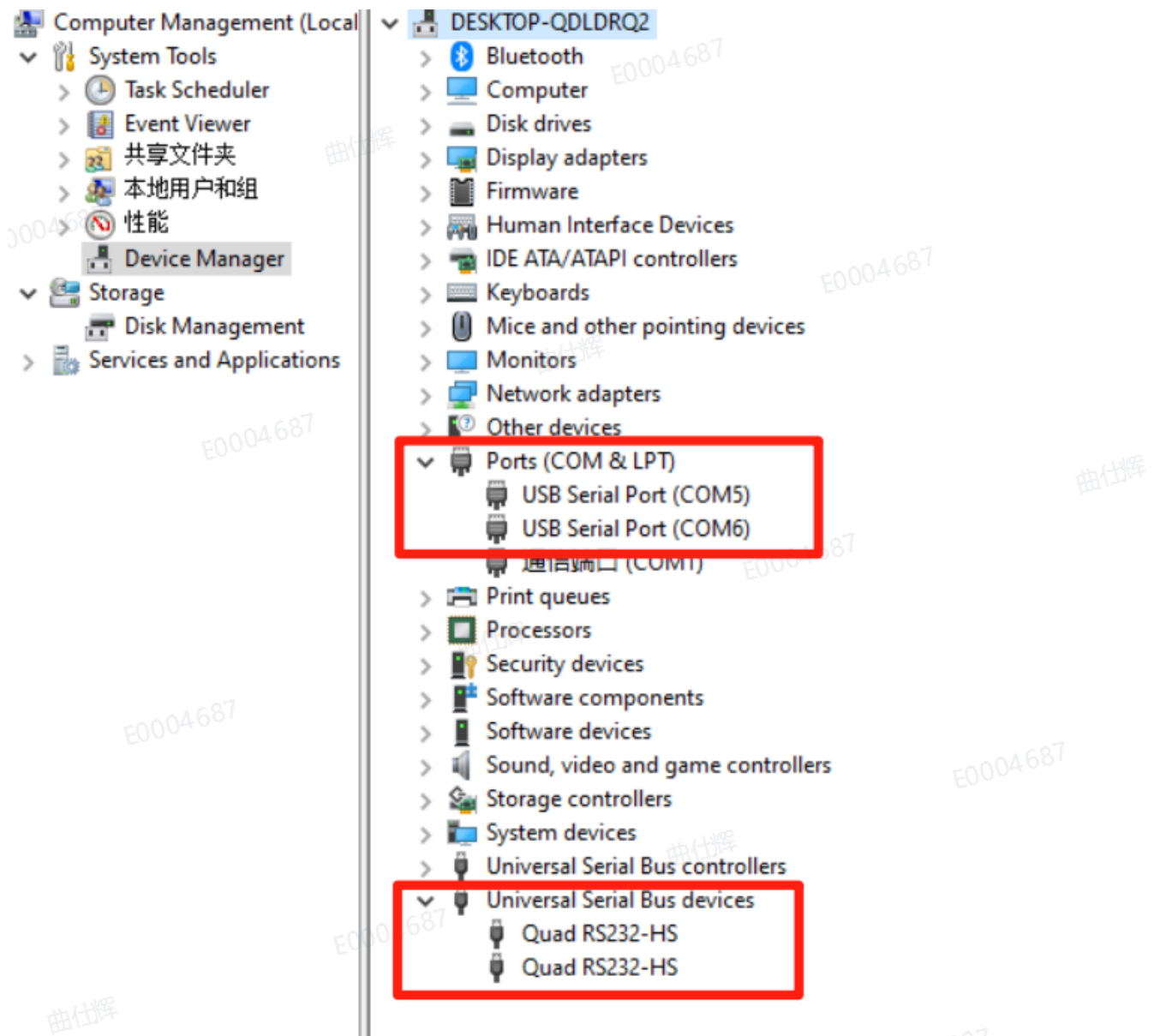


Figure 24: JTAG Devices

5.2. Fastboot Driver Installation (System Flashing)

Fastboot is used to quickly flash Debian system boot and root partition images via USB.

Note

In addition to Fastboot flashing, ESWIN also provides a USB flash drive flashing method. If you are not using Fastboot, please skip this section.

5.2.1. Preparation

1. Acquire Tools:

- [Click to download](#) the Fastboot command-line tool. Extract it and add the `platform-tools` directory to the system's `PATH` environment variable.
- [Click to download](#) the `Android USB Driver for Windows` package for later use.

2. Physical Connection:

- Use the USB Type-A cable from the accessories to connect the **USB0** port at the top of the P550 rear panel to the PC.
- Ensure the Type-C debug port is also connected to the PC to access the U-Boot command line.

Note

The P550 rear panel has two USB Type-A ports:

- **Top (USB0):** Configured as USB Device (Slave) mode in U-Boot, dedicated to Fastboot communication.
- **Bottom (USB1):** USB Host mode, used to connect peripherals like USB flash drives.

Please be sure to connect the top (USB0) port for Fastboot operations.

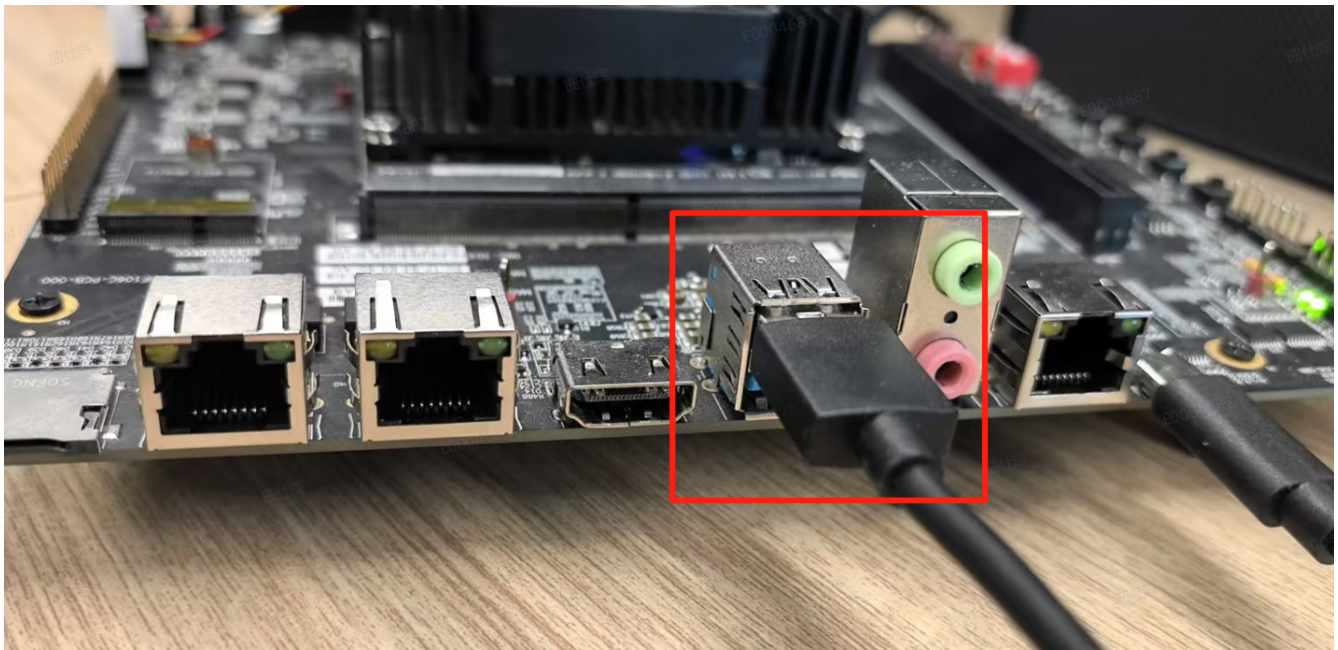


Figure 25: USB0

5.2.2. Entering Fastboot Mode and Installing the Driver

1. Boot the development board via the serial terminal (connected to SoC UART, e.g., COM5). Press Enter during the U-Boot boot countdown to enter the U-Boot command line.
2. In the U-Boot command line, enter the following command to set USB0 to Fastboot device mode. The command line will return a message similar to `generic_phy_get_bulk: no phys property`, indicating it has entered the waiting for connection state.

```
=> fastboot usb 0
generic_phy_get_bulk : no phys property
```

3. At this point, a new unknown device named `USB download gadget` will appear under “Other devices” in the PC’s “Device Manager” .

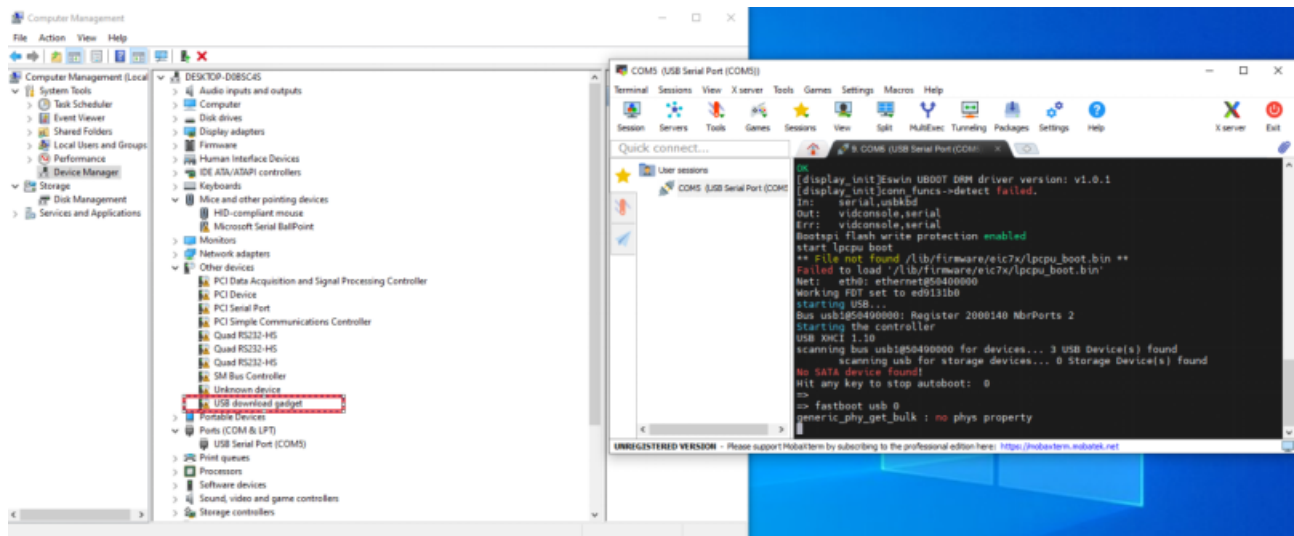


Figure 26: Fastboot Device Port

4. Right-click the **USB download gadget** device and select **Update driver**.

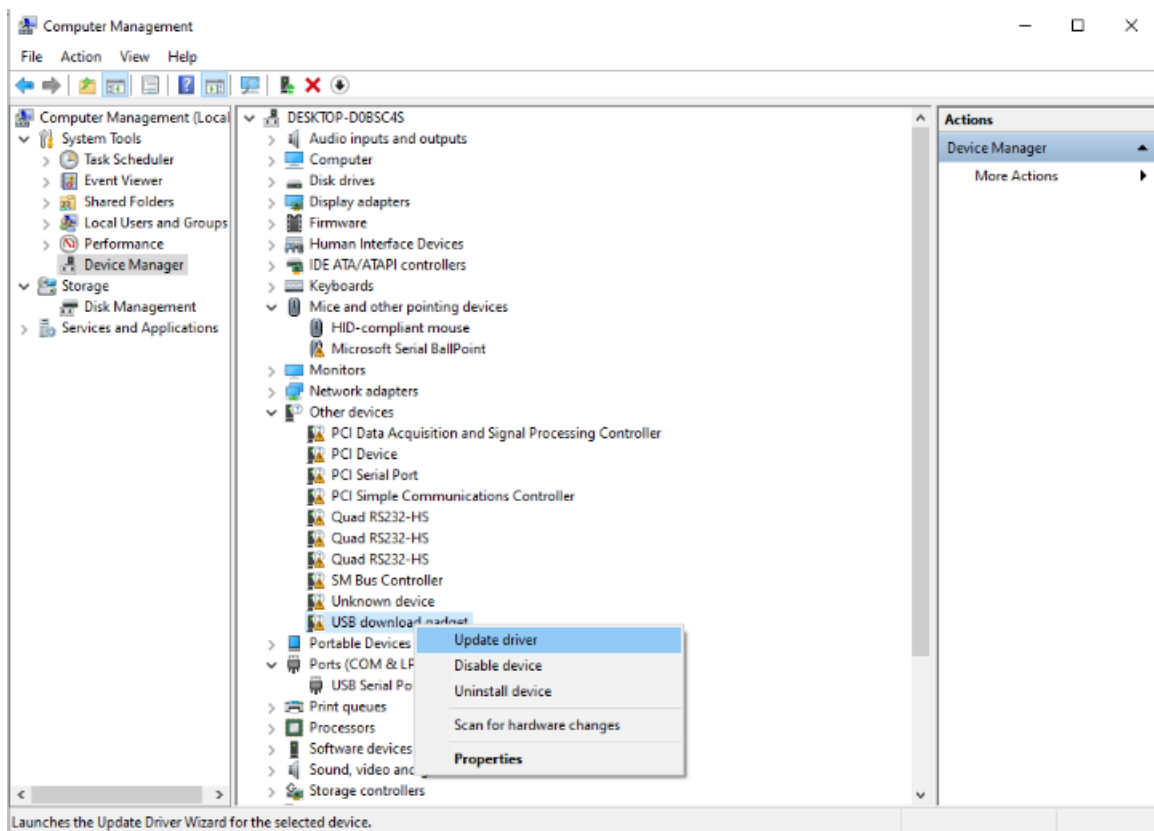


Figure 27: Fastboot Driver Update

5. In the pop-up window, select **Browse my computer for drivers**.

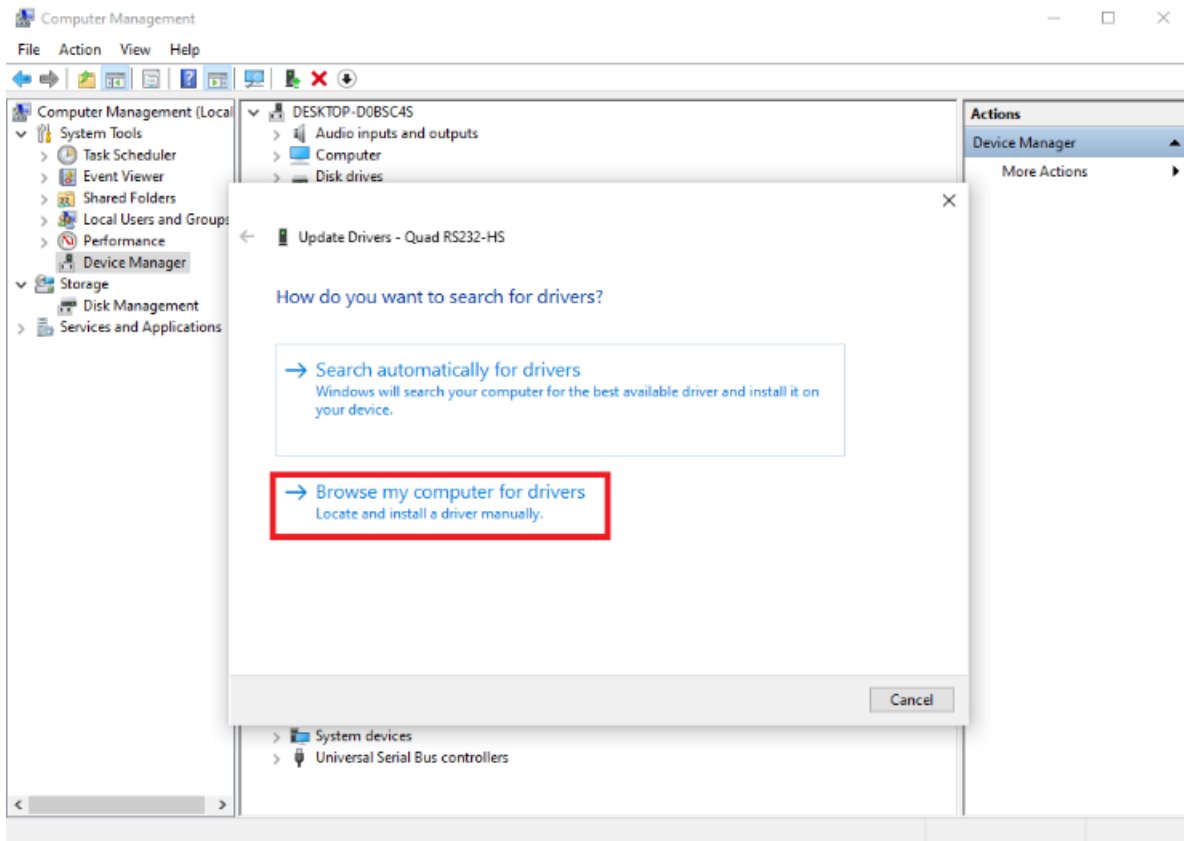


Figure 28: Fastboot Driver Browse

6. Select **Let me pick from a list of available drivers on my computer**.

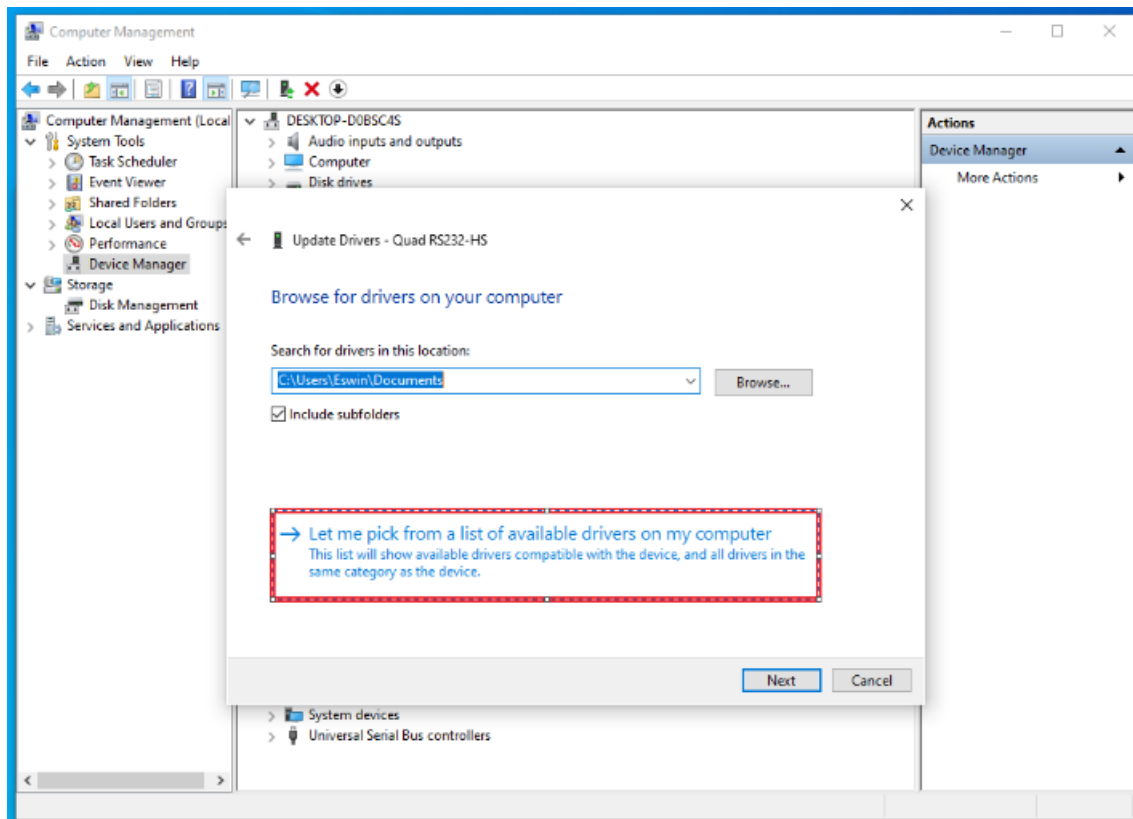


Figure 29: Fastboot Driver Select File

7. In the device type list, select **ADB Interface** and click Next.

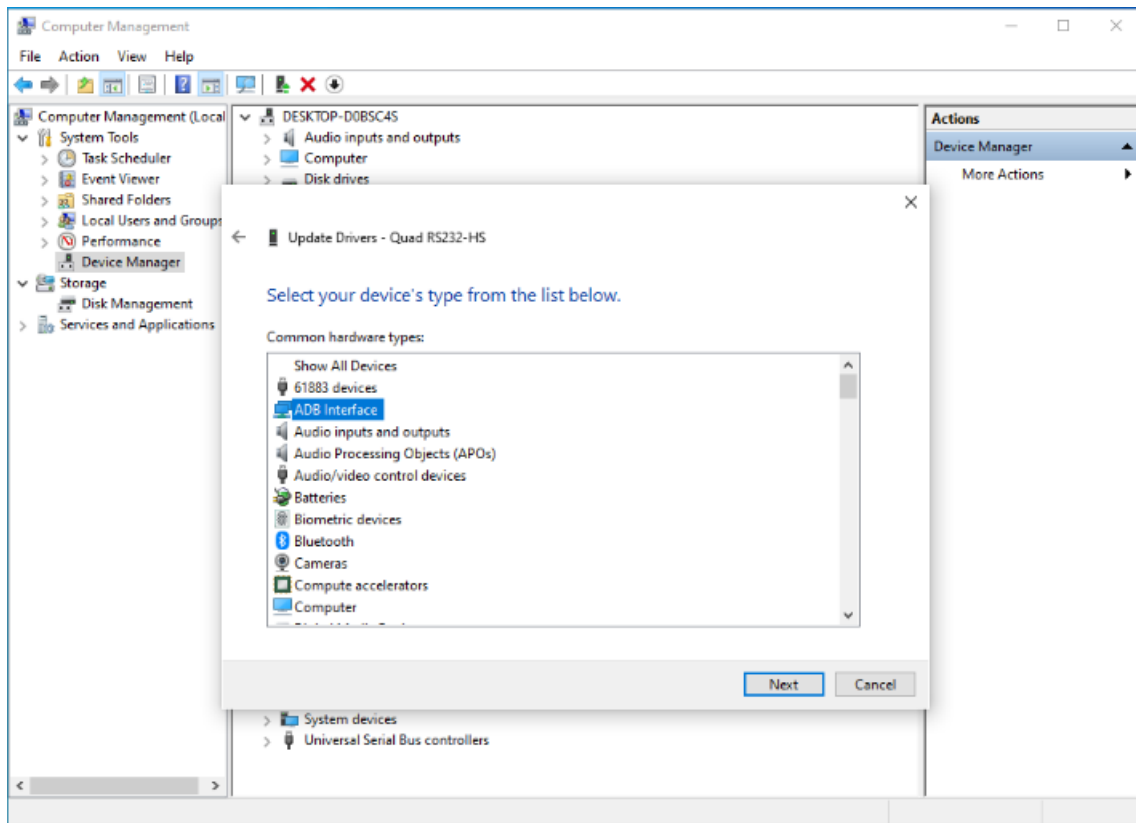


Figure 30: Fastboot Driver Select Device Type

8. Select the first driver and click Next to complete the installation.

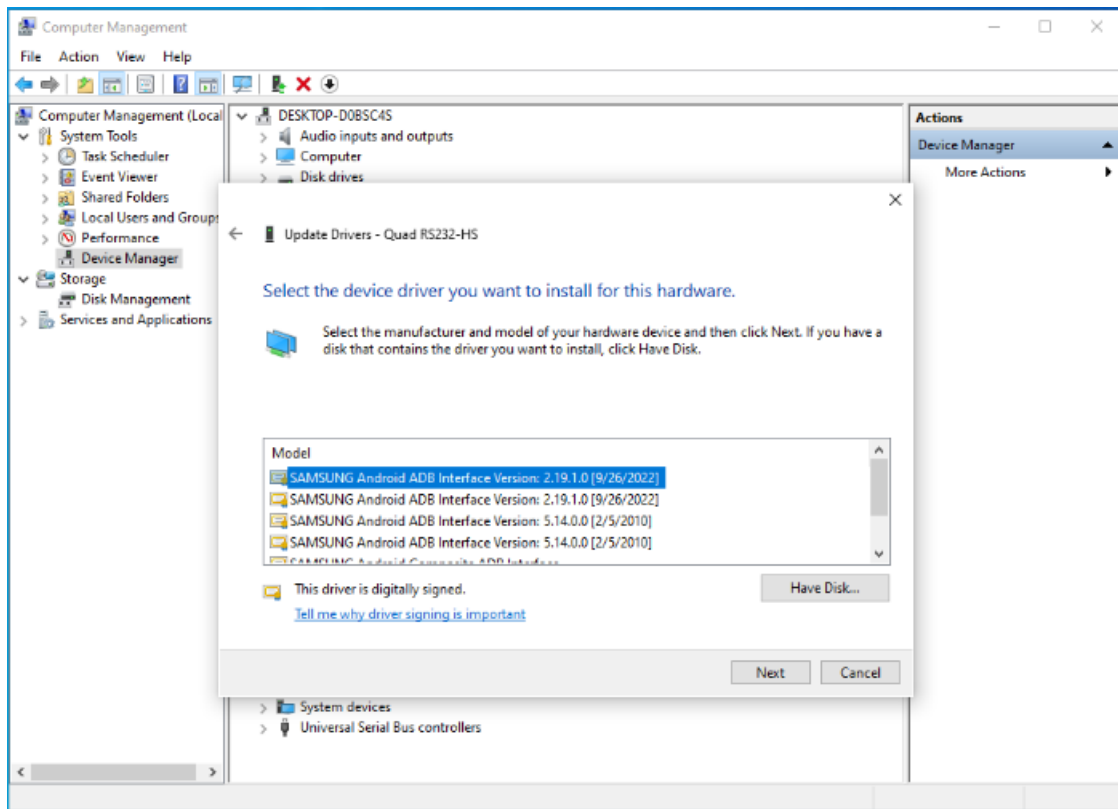


Figure 31: Fastboot Driver Select Driver

9. If the system prompts with a **Windows cannot verify the publisher of this driver** warning, select **Yes**.

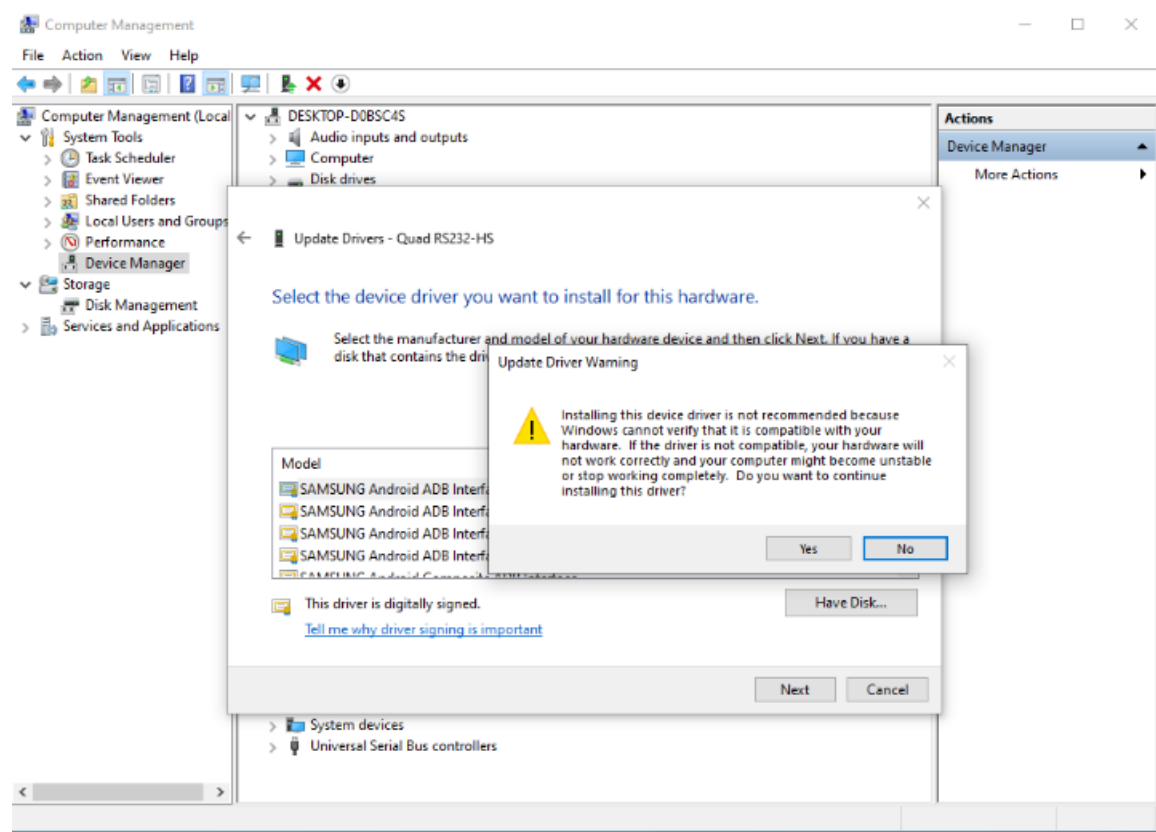


Figure 32: Fastboot Driver Confirm

5.2.3. Verifying Fastboot Connection

After the driver is successfully installed, a new device should be visible under **ADB Interface** in **Device Manager** .

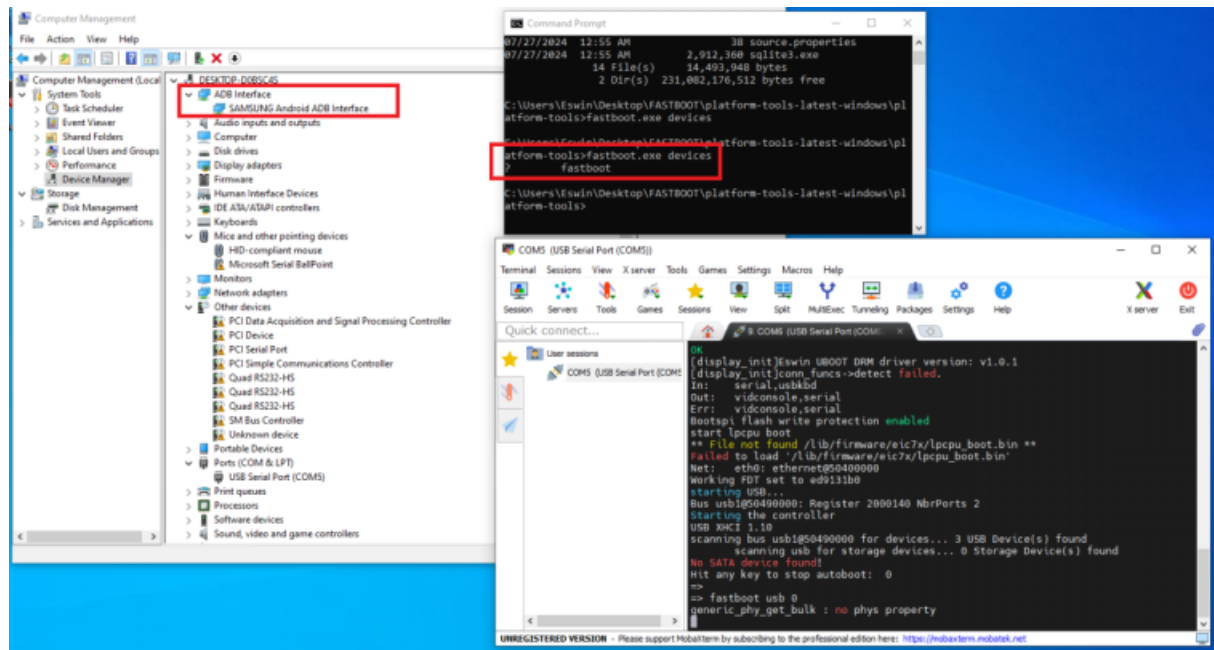


Figure 33: Fastboot Driver Refresh

1. Open Command Prompt (CMD) or PowerShell on your PC.
2. Enter the following command to check if the device is recognized:

```
fastboot devices
```

3. If the connection is successful, the command line will return the device serial number with the status `fastboot`, for example:

```
xxxxxxx fastboot
```

At this point, the Fastboot driver installation is complete, and you can begin using Fastboot commands to flash system images.

6. System Image Burning Guide

This chapter guides you on how to flash or update the system software for the P550 development board. The flashing operations primarily cover two parts: the Bootloader (U-Boot) and the Operating System (Debian/Ubuntu). Before starting any flashing operations, please ensure you have:

1. Completed all preparations in [Hardware Introduction and Usage Guide](#) (especially the heatsink installation).
2. Installed the necessary drivers as described in [Windows Driver Installation and Debugging Guide](#) based on your development environment (Windows/Linux).
3. Prepared a reliable USB Type-C data cable (for debugging and command control) and a USB Type-A data cable (Required for Fastboot flashing or USB recovery).

The P550 supports flashing via multiple methods. This chapter will detail the most commonly used approaches.

6.1. Obtaining System Image

6.1.1. Official Images & Custom Builds

ESWIN officially maintains and provides two mainstream Linux distribution images for the P550 development board:

Distribution	Description	How to Obtain
Debian	A stable, general-purpose Linux distribution suitable for most development and evaluation scenarios.	Download the pre-compiled Debian system image package from the official website .
Ubuntu 24.04	A popular Linux distribution with an active community, offering newer software packages and Long Term Support (LTS) versions.	Download the pre-compiled Ubuntu system image package from the official website .

Tip

If you need to customize the system based on specific requirements (such as kernel configuration, drivers, or software packages), please refer to the independent [BSP Quick Start Guide](#) for system compilation.

6.1.2. Detailed Image File Structure

The image downloaded from the official website is a compressed archive. After extracting it, you will obtain a set of specific image files. Understanding the function of each file is a prerequisite for correct flashing.

6.1.2.1. Debian System Image File Structure

1. Download the `P550-xxxx-debian.tar.zst` archive to your local machine.
2. Extract it using a tool that supports the zstd compression format (the following Linux command line is used as an example):

```
tar --zstd -xvf P550-xxxx-debian.tar.zst
```

3. After extraction, you will obtain a `P550/` directory containing the following 4 core image files:

File Name	Description	Key Function
bootloader_P550.bin	Bootloader (U-Boot) Main Image	The boot program that runs first after power-on. Responsible for hardware initialization and loading the kernel. Typically used to upgrade an existing functional Bootloader, flashed into SPI FLASH.
recovery_bootloader_P550.bin	Bootloader Recovery Image	Used for system repair when the Bootloader in SPI Flash is corrupted and cannot boot normally. Requires use with “USB Boot Mode”.
boot-P550-YYYYMMDD-HHMM.ext4	Linux Kernel Partition Image	Contains the Linux kernel, device tree (dtb), and initial ramdisk (initrd). Flashed into eMMC or SD card.
root-P550-YYYYMMDD-HHMM.ext4	Linux Root Filesystem Image	Contains the complete operating system files, tools, and user-space programs. Flashed into eMMC or SD card.

Tip

Since version [26.06], the naming prefix of all images has been updated from **P550** to **eic7700-hifive-premier-p550**. The table above retains the original naming for reference only; please refer to the official release for the actual file names.

6.1.2.2. Ubuntu System Image File Structure

1. Download the **P550-xxxx-ubuntu.tar.zst** archive to your local machine.
2. Extract it using a tool that supports the zstd compression format (the following Linux command line is used as an example):

```
tar --zstd -xvf P550-xxxx-ubuntu.tar.zst
```

3. After extraction, you will obtain a **P550/** directory containing the following 3 core image files:

File Name	Description	Key Function
bootloader_P550.bin	Bootloader (U-Boot) Main Image	Same as above, used to upgrade the bootloader.
recovery_bootloader_P550.bin	Bootloader Recovery Image	Same as above, used for system repair.
P550-ubuntu-24.04-preinstalled-server-riscv64_*.img	Full System Image	A unified image file containing both the Boot and Root partitions of the Ubuntu system. Flashed into eMMC or SD card.

Tip

Since version [26.06], the naming prefix of all images has been updated from **P550** to **eic7700-hifive-premier-p550**. The table above retains the original naming for reference only;

please refer to the official release for the actual file names.

Danger

Pre-upgrade Must-Read: Bootloader Pre-upgrade Guide

Background: The new version EIC7X-26.06 introduces a Bootloader A/B backup mechanism to prevent bricking during upgrades. Due to changes in the partition structure and the old Bootloader's lack of support for the new A/B switching logic, old versions (20251230 and earlier) cannot be directly overwritten using the `es_burn` command. You must manually upgrade the Bootloader first.

Operation Paths (choose one):

1. Option 1 (Recommended): Refer to [Scenario 2: Bootloader fails to boot\(USB Mode Recovery\)](#) and complete the upgrade through the repair process.
2. Option 2 (Manual Tool): Use the dedicated bootloader-tool for command-line flashing.
 - Extract the `bootloader_eic7700-hifive-premier-p550.bin` file from the image package `P550-26.06-debian.tar.zst`, and download the `bootloader-tool` file from the official website (The upgrade method for Ubuntu is the same as for Debian).
 - After booting the old system, transfer these two files to the device system directory (e.g., `/home/eswin/`).
 - Open a terminal, navigate to the file directory (e.g., `cd /home/eswin/`), and execute the following commands in order:

```
sudo ./bootloader-tool getinfo 0
sudo ./bootloader-tool upgrade bootloader_eic7700-hifive-premier-p550.bin
```

- After flashing is complete, please restart the device and enter Uboot mode.
- Once device connection is confirmed normal, you can proceed to upgrade the main system following the standard burning process.

6.2. Bootloader Flashing Guide

Warning

The Bootloader is the cornerstone of system startup. Incorrect flashing steps, image files, or operational interruptions can render the development board unbootable. Please strictly follow this guide and ensure stable power during the operation.

Debian and Ubuntu systems use the same Bootloader image, so this guide is applicable to both. Bootloader flashing is divided into two scenarios based on its current state:

- **Scenario 1: Bootloader is functional** - Upgrade the existing Bootloader from a USB drive via U-Boot command line.
- **Scenario 2: Bootloader is unbootable** - Force restore the Bootloader via “USB Boot Mode” .

Please select the corresponding section to proceed based on your actual situation.

6.2.1. Scenario 1: Bootloader is Functional (USB Upgrade)

This method applies to routine upgrades where the Bootloader is operational and can access the U-Boot command line.

6.2.1.1. Preparation

1. **Prepare USB Drive:** Find a USB drive and format it to the ext4 or FAT32 file system. The ext4 format is recommended because FAT32 has a 4GB single file size limit, which is relevant when flashing larger rootfs images later. U-Boot supports reading both formats.
2. **Place Image File:** Copy the `bootloader_P550.bin` file that needs to be upgraded to the root directory of this USB drive.

6.2.1.2. Operation Steps

1. **Connect and Boot:** Insert the prepared USB drive into the USB1 interface (the lower interface) on the back panel of the development board. Boot the development board and press Enter at the U-Boot countdown screen to enter the U-Boot command line.
2. **Identify USB Device:** Enter the following command in the U-Boot command line to initialize USB and scan for storage devices.

```
=> usb reset
```

After the command executes, please confirm that the output contains `1 Storage Device(s) found`. This indicates the USB drive has been successfully recognized. If not found, please check if the USB drive is properly inserted, if the format is correct, and if it is plugged into the correct USB port.

3. **List USB Files:** Enter the following command to view the files in the root directory of the USB drive and confirm that `bootloader_P550.bin` is present.

```
=> ls usb 0
```

Note

If the USB drive has multiple partitions and the image file is not on the first one, you can use `ls usb 0:2` to view the second partition (and so on).

4. **Load Image to Memory:** Depending on the file system format of the USB drive, use the corresponding command to load the Bootloader image to the specified memory address (`0x90000000`).

- **ext4 format USB drive:**

```
=> ext4load usb 0 0x90000000 bootloader_P550.bin
```

- **FAT32 format USB drive:**

```
=> fatload usb 0 0x90000000 bootloader_P550.bin
```

After successful loading, the terminal will display the number of bytes read and the speed.

5. **Flash Image to SPI Flash:** Execute the following critical command to write the image from memory to the onboard SPI Flash.

```
=> es_burn write 0x90000000 flash
```

This process will output detailed erasing and writing progress. Please wait until the final prompt `bootloader write OK` appears, indicating the flashing was successful.

Warning

Example of Abnormal Output:

```
=> es_burn write 0x90000000 flash
Bootspi flash write protection disabled
SF: 4096 bytes @ 0x1000 Read: OK
SF: 4096 bytes @ 0x0 Read: OK
UPDATE BOOTLOADER FAILED (Invalid image file type)
Bootspi flash write protection enabled
exit not allowed from main input shell.
```

If the above error message appears, please erase the flash first and then rewrite it. The operation steps are as follows:

- Erase command: `es_burn erase flash`
- Write command: `es_burn write 0x90000000 flash`

6. **Reboot System:** After flashing is complete, enter the following command to reboot the development board; the new Bootloader will take effect immediately.

```
=> reset
```

6.2.2. Scenario 2: Bootloader is Unbootable (USB Mode Recovery)

Warning

The method in this section is only for repairing “bricked” devices where the Bootloader is corrupted and cannot enter any command line. If the Bootloader can start normally, please prioritize the safer USB drive upgrade method described above.

When the Bootloader cannot start, you need to set the development board to USB device mode to recover directly from the host (PC).

6.2.2.1. Prerequisites

1. **Cable:** Use the USB Type-A data cable included in the accessories.
2. **Prepare Recovery Image:** Place the `recovery_bootloader_P550.bin` file in an easily accessible location on your PC.

6.2.2.2. Procedure

1. **Hardware Connection:** Use a dual USB Type-A cable to connect the USB0 interface on the upper part of the development board's back panel to the PC.
2. **Set USB Boot Mode:**
 - Set the onboard DIP switch SW1 to `1100` (USB Boot Mode). For specific setup instructions, please refer to [Boot Mode Selection](#).
 - Connect the power supply, turn on the `Main Switch`, and then press the `Power Button` to power on.
3. **Boot Development Board:** After the development board boots, the PC will recognize it as a removable disk (USB drive) named `ESWIN-2030`.

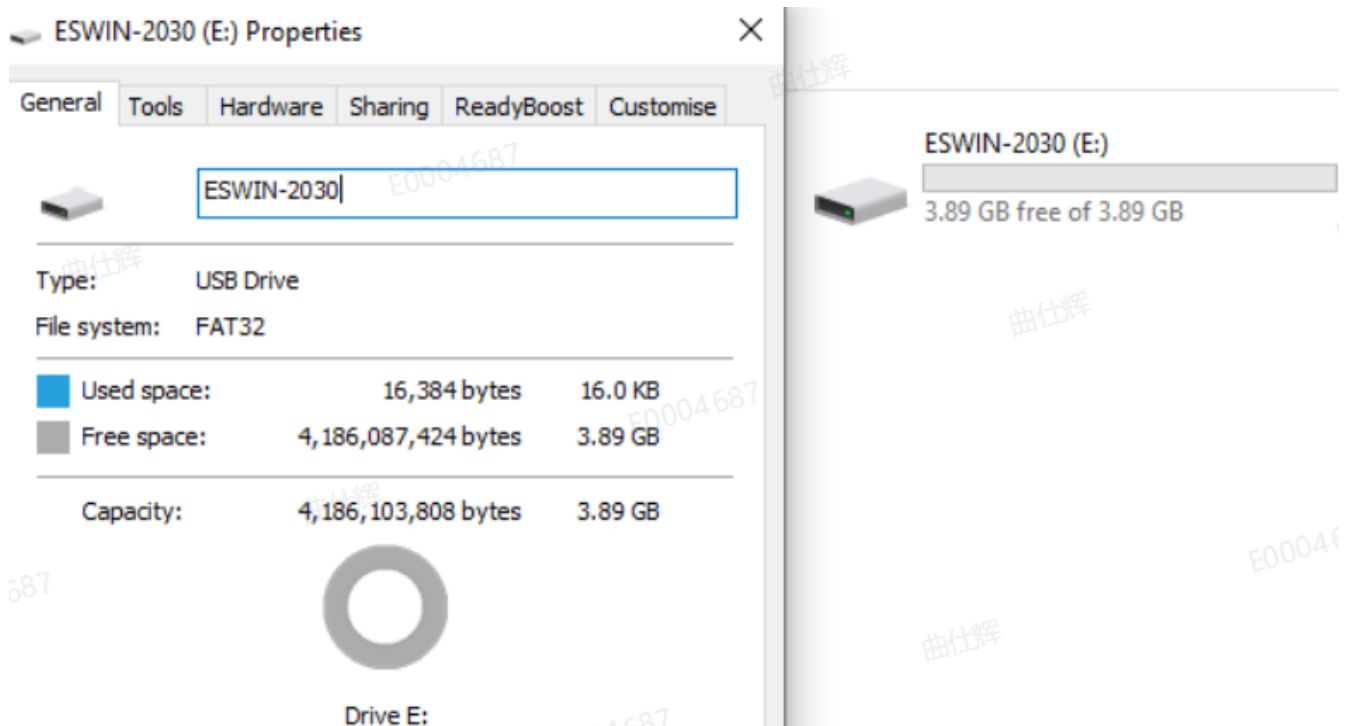


Figure 34: Simulated USB Drive Device

4. **Copy Recovery Image:** Copy or drag and drop the previously prepared `recovery_bootloader_P550.bin` file into this disk.

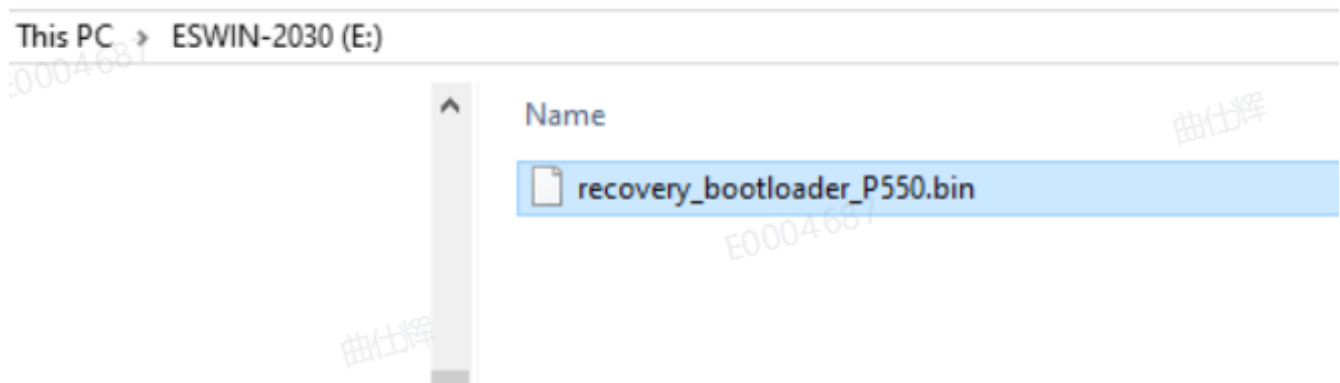


Figure 35: copy recovery_bootloader_P550.bin

5. **Wait for Automatic Recovery:** After the file copy is complete, the development board will automatically begin the recovery process. You can observe the progress in two ways:
 - **HDMI Monitor:** The screen will display an `Upgrading...` progress bar, which changes to `Upgrade Finished` upon completion.
 - **Serial Terminal:** Detailed recovery logs can be viewed via a serial tool (connected to SoC UART). The final indicator of successful recovery is the appearance of `bootloader write OK with logo` in the logs.
6. **Recovery Complete and Reboot:**
 - When the monitor indicates the upgrade is complete, or the serial logs show a successful write, the Bootloader has been restored.
 - Power off the development board (turn off the `Main Switch` or unplug the power cable).
 - Reset the DIP switch `SW1` back to `0100` (Boot from SPI Flash).

- Power on again; the development board will boot normally using the newly restored Bootloader.

Tip

Since version [26.06], the naming prefix of all images has been updated from `P550` to `eic7700-hifive-premier-p550`.

6.2.2.3. Post-Recovery Operations

Once the Bootloader is successfully restored, the development board will be able to boot normally. Next, you will need to proceed with flashing the full operating system (Debian or Ubuntu) onto the board.

6.3. Flashing Debian System Image

The Debian system image supports two mainstream flashing methods: USB Drive Flashing and Fastboot Flashing.

Tip

1. **USB Drive Flashing:** Requires only a formatted USB drive, suitable for all environments, and is especially recommended for beginners.
2. **Fastboot Flashing:** Requires driver installation and network configuration on the PC, suitable for users familiar with Fastboot tools.

6.3.1. Method 1: USB Drive Flashing Guide

This method writes the system image directly to the on-board eMMC storage from a USB drive via the U-Boot command line.

6.3.1.1. Prerequisites

1. **Prepare USB Drive:** Prepare a USB drive with a capacity of at least 16GB and format it to the ext4 file system.
2. **Obtain Image Files:** Download and extract the Debian system package from [Obtaining System Images](#) to get the `boot-P550-YYYYMMDD-HHMM.ext4` (kernel partition image) and `root-P550-YYYYMMDD-HHMM.ext4` (root file system image) files.
3. **Place Images:** Copy the two `.ext4` image files mentioned above to the root directory of the USB drive.

6.3.1.2. Operation Steps

Please strictly follow the steps below in order.

1. **Connect USB Drive and Boot:** Insert the prepared USB drive into the USB1 interface (the lower interface) on the back panel of the development board. Boot the development board and press the Enter key during the U-Boot countdown to enter the U-Boot command line.
2. **Identify USB Device:** In the U-Boot command line, enter the following command to scan for USB storage devices:

```
=> usb reset
```

After the command is executed, please confirm that the output contains **1 Storage Device(s) found** . This indicates that the USB drive has been successfully recognized. If not found, please check if the USB drive is properly inserted, if the format is correct, and if it is plugged into the wrong USB port.

3. **Verify Image Files:** Enter the following command to list the files in the root directory of the USB drive and confirm that the two system image files exist:

```
=> ls usb 0
```

Tip

If the USB drive has multiple partitions and the image file is not on the first one, you can use `ls usb 0:2` to view the second partition (and so on).

4. **Check Target Storage Partitions:** Before flashing, you need to confirm that the partition table of the on-board eMMC (`mmc 0`) is correct. Execute the following command to view the partition information:

```
=> mmc part
```

Partition Map for MMC device 0 -- Partition Type: EFI

Part	Start LBA	End LBA	Name
Attributes			
Type GUID			
Partition GUID			
1	0x00000800	0x001007ff	"boot"
attrs: 0x0000000000000000			
type: c12a7328-f81f-11d2-ba4b-00a0c93ec93b			
(system)			
guid: 44b7cb94-f58c-4ba6-bfa4-7d2dce09a3a5			
2	0x00100800	0x009007ff	"swap"
attrs: 0x0000000000000000			
type: 0657fd6d-a4ab-43c4-84e5-0933c84b4f4f			
(swap)			
guid: 5ebcaaf0-e098-43b9-beef-1f8deedd135e			
3	0x00900800	0x045007ff	"root"
attrs: 0x0000000000000000			
type: 0fc63daf-8483-4772-8e79-3d69d8477de4			
(linux)			
guid: b0f77ad6-36cd-4a99-a8c0-31d73649aa08			
4	0x04500800	0x0e8f5fde	"userdata"
attrs: 0x0000000000000000			
type: 0fc63daf-8483-4772-8e79-3d69d8477de4			
(linux)			
guid: 9407dd07-6354-4542-af49-71e32d5ddd82			

You need to specifically confirm that the output contains partition **1** labeled `boot` and partition **3** labeled `root` . Correct partitions are a prerequisite for successful flashing.

Warning

- **If the partition information is correct:** Please proceed directly to Step 5.
- **If partitions are missing or incorrect:** You must perform partitioning first; otherwise,

flashing will fail or the system will be unable to boot. Enter the following command to recreate the standard GPT partition table:

```
=> run gpt_partition
```

After seeing the `Writing GPT: success!` prompt, be sure to execute the `mmc part` command again to confirm that the `boot(1)` and `root(3)` partitions have been correctly created.

5. **Flash Boot Partition Image:** Write the kernel image from the USB drive to the `boot` partition of the eMMC (`mmc 0:1`). This process will take several minutes; please wait patiently until it is complete.

```
=> es_fs update usb 0 boot-P550-YYYYMMDD-HHMM.ext4 mmc 0:1
```

Success Indicator: `mmc has been successfully written in mmc 0:1` should be displayed at the end of the command output.

6. **Flash Root Partition Image:** Write the root file system image from the USB drive to the `root` partition of the eMMC (`mmc 0:3`). This is the largest image and will take the longest time; do not interrupt the process.

```
=> es_fs update usb 0 root-P550-YYYYMMDD-HHMM.ext4 mmc 0:3
```

Success Indicator: `mmc has been successfully written in mmc 0:3` should be displayed at the end of the command output.

7. **Complete and Reboot:** After both images are successfully flashed, enter the following command to reboot the development board:

```
=> reset
```

The development board will boot from the newly flashed Debian system. The first boot will involve system initialization, which may take a slightly longer time. Please wait patiently until the login prompt appears.

6.3.1.3. Troubleshooting

1. **Error during flashing:** Please re-check if the USB drive is securely connected, if the image files are intact, and if the eMMC partitions (Step 4) are correct.
2. **System fails to boot:** Please confirm that you have strictly followed Steps 5 and 6, flashing the images to the correct partitions (`mmc 0:1` and `mmc 0:3`). You can check the boot logs via the serial port to determine the failure stage.

6.3.2. Method 2: Fastboot Flashing Guide

This method uses a USB connection to quickly flash images from a PC host using the Fastboot tool.

6.3.2.1. Prerequisites

1. **PC Drivers Installed:** Fastboot drivers must be installed on the PC(see [Driver Installation Guide](#)).
2. **Cable Connections:** The development board and PC must be connected via two cables:

- **USB Type-C cable:** For the serial console (to enter U-Boot).
 - **Dual-headed USB Type-A cable:** For Fastboot data transfer (connect to the development board's USB0 port).
3. **Obtain Image Files:** Same as USB flashing, prepare the `boot-P550-...ext4` and `root-P550-...ext4` image files on the PC.
 4. **Prepare PC Environment:** Ensure the PC has the `fastboot` command-line tool installed and that its directory has been added to the system `PATH` environment variable.

6.3.2.2. Operation Steps

1. **Enter U-Boot and Check Partitions:**

- Boot the development board via the serial terminal (e.g., connected to COM5). Press Enter during the U-Boot countdown to enter the command line.
- Execute the `mmc part` command to check and confirm that the `boot(1)` and `root(3)` partitions exist and are correct. If repair is needed, execute the `run gpt_partition` command.

2. **Start Development Board Fastboot Mode:** In the U-Boot command line, enter the following command to set USB0 as the Fastboot device:

```
=> fastboot usb 0
```

The command line will return prompts such as `generic_phy_get_bulk: no phys property`, indicating that the development board has entered a waiting-for-connection state.

3. **Verify Fastboot Connection on PC:** Open the Command Prompt (CMD) or terminal on your PC and execute:

```
fastboot devices
```

If the connection is normal, it will return the device serial number with the status `fastboot`.

4. **Flash Boot Partition Image:** In the PC command line, navigate to the directory where the image files are located and execute:

```
fastboot flash boot boot-P550-YYYYMMDD-HHMM.ext4
```

Wait for the flashing to complete; the command line will display the transfer speed and an "OKAY" status.

5. **Flash Root Partition Image:** In the PC command line, continue to execute:

```
fastboot flash root root-P550-YYYYMMDD-HHMM.ext4
```

This is the largest image and will take a long time; do not interrupt the process.

6. **Finish and Reboot:** After both images are flashed, execute the reboot command in the PC command line:

```
fastboot reboot
```

The development board will automatically reboot and start from the newly flashed system.

Tip

On Linux, please use `sudo fastboot` to execute commands.

6.3.2.3. Advanced Feature: Network (UDP) Fastboot

The Fastboot tool supports flashing via the Ethernet UDP protocol, which can be faster and does not require a USB data cable.

- Connect the development board's Ethernet ETH0 port to the network, ensuring it is on the same local area network (LAN) as the PC host and that a DHCP service is available.
- **In U-Boot:** Use the `fastboot udp 0` command to start the network Fastboot service and assign an IP address to the development board.
- **On PC:** Use `fastboot -s udp:<Dev_Board_IP> flash` to specify the target device for flashing.
- For specific configuration steps, please contact ESWIN technical support (FAE).

6.4. Flashing Ubuntu System Image

The Ubuntu system image uses a single full-disk image file. The flashing process is simpler compared to Debian. It involves writing the Ubuntu full-disk image directly from a USB drive to the onboard eMMC storage via U-Boot command line.

6.4.1. Prerequisites

1. **Prepare USB Drive:** Prepare a USB drive with a capacity of at least 16GB and format it to the ext4 file system.
2. **Obtain Image File:** Download and extract the system package from [Obtaining System Images](#) to get the `P550-ubuntu-24.04-preinstalled-server-riscv64_*.img` file (hereinafter referred to as `ubuntu-*.img`).
3. **Place Image:** Copy the aforementioned `.img` full-disk image file to the root directory of the USB drive.

6.4.2. Operation Steps

Please strictly follow the steps below in order.

1. **Connect USB Drive and Boot:** Insert the prepared USB drive into the USB1 interface (the lower interface) on the back panel of the development board. Power on the development board and press Enter at the U-Boot countdown screen to enter the U-Boot command line.
2. **Initialize and Identify USB Drive:** Enter the following command in the U-Boot command line to scan for USB storage devices:

```
=> usb reset
```

After the command is executed, please confirm that `1 Storage Device(s) found` is included in the output informatio. This indicates that the USB drive has been successfully recognized. If not found, please check if the USB drive is properly inserted, if the format is correct, and if it is plugged into the correct USB port.

3. **Verify Image File:** Enter the following command to list the files in the root directory of the USB drive and confirm that the Ubuntu full-disk image file exists:

```
=> ls usb 0
<DIR>    4096 .
<DIR>    4096 ..
<DIR> 16384 lost+found
      8523011584 P550-ubuntu-24.04-preinstalled-server-riscv64_20260313_1344_68.img
```

Tip

If the USB drive has multiple partitions and the image file is not located in the first partition, you can use `ls usb 0:2` to view the second partition (and so on).

4. **Flash Full Disk Image:** Execute the `es_fs write` command to write the image file to the target storage device byte-by-byte, automatically overwriting existing partitions and data on it.

```
=> setenv stdout serial
=> es_fs write usb 0 P550-ubuntu-24.04-preinstalled-server-riscv64_20260313_1344_68.img mmc 0
Write progress: 100%:+++++
P550-ubuntu-24.04-preinstalled-server-riscv64_20260313_1344_68.img has been successfully written in
↳ mmc 0s
```

Warning

1. In U-Boot, use `setenv stdout serial` to solve the problem of pasted commands being truncated.
2. When executing `es_fs write` for flashing, please replace `xxx.img` in the example command with the actual image file name.
3. This process will take a long time (depending on the image size and USB drive speed). Do not power off or perform any operations until the command execution is complete.

5. **Completion and Reboot:** After the flashing command executes successfully, enter the following command to reboot the development board:

```
=> reset
```

The development board will boot from the newly flashed Ubuntu system.

6.4.2.1. First Login and Password

After logging into the system for the first time, you will be required to change the default password. Depending on your operational environment, you can follow either of the methods below:

Method 1: Change Password via the System Login Interface

1. Connect an HDMI display and USB keyboard, system boot up and enter the login interface.
2. Use the default username in the login window.
3. Enter the default password: `ubuntu` (in lowercase).
4. After logging in, a prompt will appear below the password input box: You are required to change your password immediately (administrator enforced)
5. As prompted, enter the current password again: `ubuntu`.
6. Enter a new password (the new password cannot be the same as the old one) and retype it to confirm.
7. After successfully changing the password, the system will proceed to the desktop as normal.

Method 2: Change Password via Serial Port Login

1. Connect to the device using a serial port tool and access the login interface.

2. Enter the username: ubuntu.
3. Enter the default password: ubuntu.
4. The system will prompt you to change the password: You are required to change your password immediately (administrator enforced)
5. As prompted, enter the current password: ubuntu.
6. Set a new password and enter it again to confirm (the new password cannot be the same as the old one).
7. Upon successful modification, the system will display the welcome interface: Welcome to Ubuntu

```
ubuntu login: ubuntu # Enter username ubuntu
Password:          # Enter password ubuntu
You are required to change your password immediately (administrator enforced).
Changing password for ubuntu.
Current password:  # Enter password again ubuntu
New password:     # Enter new password
Retype new password: # Re-enter new password
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.115-2025-eic7700 riscv64)
```

Tip

Special scenario: If you are using an Ubuntu image pre-installed or modified by ESWIN, the default password may have been changed. In this case, try logging in with username `ubuntu` and password `Eswin8888888`.

7. Frequently Asked Questions (FAQ)

This section summarizes common issues and their solutions when using the P550 development board. Please refer to this before use or when encountering problems.

7.1. System Boot and U-Boot

Q1: How do I familiarize myself with U-Boot commands and functions?

A: In the U-Boot command line, you can get help at any time by:

- Typing `help` or `?` to view a list of all available commands.
- Typing `help <command_name>` to view detailed usage instructions for a specific command (e.g., `help mmc`).

Q2: What should I do if characters are truncated or lost when pasting long commands via serial port?

A: This is a default limitation of the U-Boot serial input buffer. It can be resolved by modifying environment variables:

```
=> setenv stdout serial
=> saveenv
```

Warning

This setting will make all U-Boot output appear only on the serial terminal, and no longer on the HDMI display. To restore HDMI output, execute `setenv stdout vidconsole,serial` and `saveenv`.

Q3: Does U-Boot support USB device hot-plugging?

A: Yes. After inserting a USB drive, you need to execute the `usb reset` command in the U-Boot command line to rescan and recognize the device.

Q4: Does U-Boot support SD card device hot-plugging?

A: No. The SD card must be inserted before the system powers on; otherwise, it cannot be recognized.

7.2. Network Connection

Q5: The development board has no network connection, cannot be pinged, or cannot obtain an IP address. What should I do?

A: Please troubleshoot in the following order:

1. **Physical Connection:** Confirm the network cable is correctly connected to the development board (SoC Gigabit Ethernet port) and the router/switch.
2. **MAC Address Check:** Execute `print ethaddr` in the U-Boot command line to see if the MAC address is correctly set. The output should be in a format like `ethaddr=xx:xx:xx:xx:xx:xx`.
3. **Set MAC Address:** If `ethaddr` is empty or needs modification, execute the following commands (replace the example address with your valid MAC):

```
=> env default -a
=> env set ethaddr aa:bb:cc:11:22:33
=> saveenv
=> reset
```

7.3. System Flashing and Partitioning

Q6: After flashing, the system reports an error on boot or fails to start.

A: This is usually because environment variables left by an old version of U-Boot are incompatible with the new system. Please try the following steps to reset environment variables after flashing the new Bootloader:

```
=> env default -a
=> saveenv
=> reset
```

Warning

`env default -a` will clear all manually set environment variables you configured previously (such as MAC address, boot arguments, etc.). Please use with caution and reconfigure after execution.

Q7: Flashing prompts “partition does not exist” or “partition is too small” .

A: This indicates that the partition table on the target storage device (eMMC) is incorrect or missing. Please perform the following operations in order:

1. **Confirm Bootloader is functional:** Ensure the correct `bootloader_P550.bin` has been flashed and booted.
2. **Clear old environment variables:** Execute `env default -a` and `saveenv` (see Q6).
3. **Re-partition:** Execute the standard partitioning command:

```
=> run gpt_partition
```

After seeing the `Writing GPT: success!` prompt, you can try flashing the system image again.

Q8: The default root partition size is insufficient. How do I adjust and re-partition?

A: You need to customize the partition table parameters and then recreate the partitions.

1. View current partition definition: `=> print partitions`
2. Edit the variable: `=> edit partitions`, or directly use `env set partitions "...[your partition table string]..."` to set it.
3. Apply the new partition table: `=> run gpt_partition`

Q9: How do I confirm that the flashed system image (Bootloader/Kernel) has been successfully updated?

A: Observe the serial log during system boot:

- **Bootloader:** During the early stage of U-Boot startup, a line of information will display the U-Boot compilation date and time. Compare this time with the compilation time of the image you flashed.
- **Linux Kernel:** During kernel decompression and startup, it will also print its compilation timestamp. Comparing this time confirms whether the kernel image has been updated.

Q10: How do I partition an SD card or SATA hard drive?

A: Use the `gpt write` command, but specify the correct device identifier.

- **For on-board eMMC (mmc 0):** `gpt write mmc 0 $partitions`
- **For SD card (mmc 1):** First switch devices `mmc dev 1`, then execute `gpt write mmc 1 $partitions`.
- **For SATA devices:** Replace `mmc` with `sata`, e.g., `gpt write sata 0 $partitions`.

Warning

⚠ Destructive Operation WARNING: The `gpt write` command will permanently overwrite all partition tables and data on the target device. Please ensure the device identifier is correct and back up all important data before proceeding.